

Traceability Matrix

REMEMBER GORDON! – Rulebook $\rightleftharpoons$ Implementation Mapping

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Overview

Implemented	Descriptive	Implicit	Out-of-scope
81	10	5	13

Total mappings: 109 · Total impl sites: 245

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0/2 implemented

§1 – Introduction

§1.1 – General Comments

descriptive

manual page 1

General Comments

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is a simulation of the final battle in Great Britain’s two-year campaign to reassert her presence in the Sudan (1896–1898). Fought September 2nd, 1898, Omdurman finally broke the back of the fanatical Dervish rebellion and gained Britain a million square miles of desolate territory and two million impoverished subjects. With two players, one assumes the role of Herbert Kitchener, Sirdar (CIC) of the Anglo-Egyptian army; the other player becomes the Khalifa, Abdullah the Taiasha, absolute ruler of the Dervishes. The game is also suited for multi-player participation, with each player assuming command of one or more Dervish tribes or Anglo-Egyptian brigades.

While “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is not, strictly speaking, a beginner’s game, the mechanics of play should be familiar to players of modest experience. It is suggested that the bonus game, FALL OF KHARTOUM, and the historical scenario be played first to familiarize players with the game system prior
(see manual for full text)

§1.2 – Game Scale

descriptive

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Game Scale

Each hexagon of the mapsheet represents approximately 400–440 yards of real terrain and each day turn is the equivalent of two hours of real time. Each counter of infantry and cavalry represents between 400 and 700 men, and each of the gunboats present at the battle has its own counter. The upper echelon of command is represented by individual leader counters for the Anglo-Egyptian force; and leaders plus their retinues for the Dervish army.

 3/6 implemented

§2 – Game Components

§2.1 – The Game Maps

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The Game Maps

The Omdurman battle map represents approximately 100 square miles of real territory and includes the area north of Omdurman in which the historical battle took place as well as the dominant terrain features that influenced the course of the battle. Note that the mapsheet also contains the Turn Record Track, Turn Sequence, and Terrain Effects Chart at the top; and the Combat Tables and Howitzer Fire Scattergram in the lower right corner. The large letters “A”, “D”, “Y”, etc. are set-up hexes for the historical scenario only (9.2) and should be ignored in the campaign game. Similarly, the hexsides of the Zariba exist only in the historical scenario and should be considered clear terrain in the campaign game. Note, however, that the Anglo-Egyptian player may “construct” the Zariba in the campaign game if desired (see 5.3). All full hexes of the Omdurman game map are playable, including the seven hexes of the Howitzer Fire Scattergram.

The mini-map for the bonus game, FALL OF KHARTOUM, represents that city as it appeared in 1885. The portion (see manual for full text)

See also: §5.3, §9.2

§2.2 – Play Aids

descriptive

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Play Aids

Certain charts and tables are needed to play the game. The Terrain Effects Chart lists all terrain found on the mapsheet and the effect of each type on movement and combat. The Combat Tables describe the range effects on various weapon types and includes the Combat Results Table. Also note the Line of Sight Table on the back of this rulebook. It tells players when certain terrain types block line of sight, thus preventing direct fire attacks on enemy units. Players should become familiar with these various charts and tables prior to the beginning of play.

(see manual for full text)

§2.3 – The Units

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The Units

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:807 GH:omdurman-types/src/lib.rs:807	UnitKind	<pre>805 /// `Some(UnitKind::Marker)` or `None`. 806 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 807 pub enum UnitKind { 808 /// Foot infantry (§2.3): fire / melee / movement. 809 Infantry { fire: i32, melee: i32, movement: i32 },</pre>
omdurman-rules/src/lib.rs:788	UnitProfile	<pre>786 /// print no melee value). 787 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]</pre>

GH:omdurman-rules/src/lib.rs:788		<pre> 788 pub struct UnitProfile { 789 pub kind: UnitKind, 790 pub identity: UnitIdentity, </pre>
omdurman-rules/src/lib.rs:16 GH:omdurman-rules/src/lib.rs:16	BrigadeId	<pre> 14 15 use omdurman_types::{ 16 BrigadeId, BrigadeNationality, DayNight, DervishTribe, Faction, HexCoord, HexsideRef, Player, 17 SetupLetter, UnitKind, 18 }; </pre>
omdurman-app/src/browser.rs:12 GH:omdurman-app/src/browser.rs:12	SpriteAnnotation	<pre> 10 /// Browser-local sprite annotation (omdurman-types no longer carries this). 11 #[derive(Clone, Debug)] 12 pub struct SpriteAnnotation { 13 pub color: SpriteColor, 14 pub faction: Option<Faction>, </pre>

§2.4 – Game Parts Inventory

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Game Parts Inventory

Your complete copy of “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN includes:

- One 22 × 28 Battle of Omdurman mapsheet
- One 8½ × 11 bonus game: FALL OF KHARTOUM mapsheet
- One Rules Booklet
- One die-cut Unit Counter Sheet
- One Campaign Game Order of Appearance Card
- One ten-sided die
- One game box

§2.31 – Dervish weapon types

implemented

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Dervish artillery, gunboats, and forts fire on the “artillery” line of the Dervish Range Effects Table; Jehadia and Danagla units fire on the “rifles” line as does the Isa Zachneih unit. All other Dervish units (including leaders) are armed with spears and swords.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:513 GH:omdurman-rules/src/lib.rs:513	WeaponClass	<pre> 511 /// enum so a "spear" unit cannot accidentally fire on the "Howitzer" line. 512 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 513 pub enum WeaponClass { 514 /// Dervish spears and swords -- no ranged fire at all. 515 Melee, </pre>
omdurman-rules/src/unit_profiles.rs:465 GH:omdurman-rules/src/unit_profiles.rs:465	dervish_tribe	<pre> 463 } 464 465 fn dervish_tribe(tribe: DervishTribe) -> Option<Classification> { 466 // §2.31: "Jehadia and Danagla units fire on the 'rifles' line as does the 467 // Isa Zachneih unit. All other Dervish units (including leaders) are armed </pre>

omdurman-rules/src/unit_profiles.rs:277 GH:omdurman-rules/src/unit_profiles.rs:277	khalifa_abdullah	<pre> 275 /// battle (\$9.322). All three are interchangeable, so they share the 276 /// `DervishArtillery` identity. 277 fn khalifa_abdullah(col: u32, row: u32) -> Option<Classification> { 278 let artillery = { 279 Some(Classification { </pre>
omdurman-rules/src/lib.rs:1005 GH:omdurman-rules/src/lib.rs:1005	HowitzerResolution	<pre> 1003 /// roll on the Howitzer Fire Scattergram (\$6.64). 1004 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1005 pub struct HowitzerResolution { 1006 pub combat_results_table_roll: DieRoll, 1007 pub impact_roll: DieRoll, </pre>

§2.32 – Anglo-Egyptian weapon types

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All Anglo-Egyptian units (except gunboats, Maxims, artillery, and leaders) are armed with rifles. Maxims fire on the “Maxims” line of the Anglo-Egyptian Range Effects Table, and artillery and old gunboats fire on the “Artillery” line. New type (named) gunboats may fire on the “Howitzer”, “Artillery”, and “Maxims” lines of the Range Effects Table. (See 6.52 for the fire capabilities of the “Friendlies”).

****Sample Dervish Units**** (printed on counters): combat unit (Combat / Melee / Movement values, plus Tribe identifier); Leader (e.g. OSMAN DIGNA); Camel unit (e.g. Danagla, 4-6-12); Fort.

****Sample Anglo-Egyptian Units**** (printed on counters): Cavalry (e.g. 21 Lancers); Artillery (e.g. 32 Battery); Old Gunboat (e.g. LORD KITCHENER, 0-0-15); New Gunboat — named (e.g. Sultan, with artillery and howitzer factor, plus movement downstream / movement upstream values); Maxim Guns (fire twice per turn); Infantry (Fire Combat Factor / Melee / Movement, plus Battalion ID and Brigade ID — e.g. “2B” = 2nd British Brigade, “3E” = 3rd Egyptian Brigade).

(see manual for full text)

See also: §6.52

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:467 GH:omdurman-rules/src/lib.rs:467	GunboatId	<pre> 465 /// fire; "old" gunboats do not (rulebook §2.32). 466 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 467 pub enum GunboatId { 468 /// One of the five new-type named gunboats with howitzer capability. 469 Named(NamedGunboat), </pre>
omdurman-rules/src/lib.rs:486 GH:omdurman-rules/src/lib.rs:486	NamedGunboat	<pre> 484 /// The five named gunboats with howitzer capability (rulebook §6.64, §2.32). 485 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 486 pub enum NamedGunboat { 487 Sultan, 488 Melik, </pre>
omdurman-rules/src/lib.rs:499 GH:omdurman-rules/src/lib.rs:499	OldGunboat	<pre> 497 /// in the Maxim Second Fire and Howitzer subphase (§6.42). 498 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 499 pub enum OldGunboat { 500 LordKitchener, 501 Tamai, </pre>
omdurman-rules/src/lib.rs:471 GH:omdurman-rules/src/lib.rs:471	GunboatId::Old	<pre> 469 Named(NamedGunboat), 470 /// An old-style gunboat -- no howitzer fire (§2.32). 471 Old(OldGunboat), 472 /// A Dervish gunboat (§9.111, §10.14). 473 DervishGunboat(u8), </pre>

omdurman-rules/src/lib.rs:473 GH:omdurman-rules/src/lib.rs:473	GunboatId::Dervish-Gunboat	471 Old(OldGunboat), 472 /// A Dervish gunboat (§9.111, §10.14). 473 DervishGunboat(u8), 474 } 475
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Covered by tests: [old_gunboat_lacks_howitz](#) [old_gunboat_rejected_from_howitz_subphase](#)

0/1 implemented

§3 – Getting Started

§3 – Getting Started

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Getting Started

Spread out the mapsheet on a table. It should lie flat if you backfold it against the scored lines. The Dervish player should sit next to the west edge of the map and the Anglo-Egyptian player opposite him on the east edge. Read through the rules once, looking over the various charts as they are referred to in the various sections. Next, select a scenario and punch out only those unit counters needed to play. Later on, the rest of the unit counters should be punched out, sorted and stored by unit type.

(see manual for full text)

■ 1/1 implemented

§4 – Turn Sequence

§4 – Turn Sequence

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Turn Sequence

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is played in “Game Turns”, each of which has two “Player Turns”. The player moving first will vary according to the scenario being played. In the Campaign Game, for example, the Anglo-Egyptian player moves first.

A) Anglo-Egyptian Player Turn:

1. Anglo-Egyptian Movement Phase
2. Fire Combat Phase a. Dervish Defensive Fire b. Anglo-Egyptian Offensive Fire
 1. Direct Fire Subphase
 2. Maxim Second Fire and Howitzer Fire Subphase

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:283 GH:omdurman-rules/src/lib.rs:283	GameTurnIndex	<pre>281 /// One-based Game Turn index (1, 2, ... up to the 282 #[derive(Serialize, Deserialize, Clone, Copy, 283 PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 284 pub struct GameTurnIndex(u8); 285 impl GameTurnIndex {</pre>
omdurman-rules/src/lib.rs:313 GH:omdurman-rules/src/lib.rs:313	Phase	<pre>311 /// etc. 312 #[derive(Serialize, Deserialize, Clone, Copy, 313 PartialEq, Eq, Debug, Default)] 314 pub enum Phase { 315 /// Pre-game deployment (\$9.2/\$9.3/\$10): fixed units are placed, each side 316 /// deploys its order of battle within its legal zone, and river</pre>
omdurman-rules/src/effects.rs:679 GH:omdurman-rules/src/effects.rs:679	GameState	<pre>677 678 // ----- 679 // 3) GameState -- authoritative mutable snapshot 680 // ----- 681 </pre>
omdurman-rules/src/effects.rs:800 GH:omdurman-rules/src/effects.rs:800	GameState::new	<pre>798 pub attack: MeleeAttack, 799 pub attacker_roll: DieRoll, 800 pub defender_roll: DieRoll, 801 } 802 </pre>
omdurman-rules/src/effects.rs:45 GH:omdurman-rules/src/effects.rs:45	AdvancePhase	<pre>43 // -- Turn / phase flow ----- 44 /// Advance to the next phase (or next player-turn if melee is done) (rulebook §4). 45 AdvancePhase, 46 47 // -- Movement -----</pre>
omdurman-rules/src/effects.rs:2257 GH:omdurman-	advance_phase	<pre>2255 // 5) Phase advancement 2256 // -----</pre>

rules/src/ effects.rs:2257		<pre> 2257 2258 /// Advance the game state to the next phase (rulebook 2259 pub fn advance_phase(state: &mut GameState) -> 2260 Result<(), RuleError> { </pre>
omdurman- rules/src/ effects.rs:2330 GH:omdurman- rules/src/ effects.rs:2330	end_player_turn	<pre> 2328 Ok(()) 2329 } 2330 2331 /// End the current player's turn: recover disrupted 2332 units, switch active player, advance turn index (rulebook 2333 §4). 2334 pub fn end_player_turn(state: &mut GameState) -> 2335 Result<(), RuleError> { </pre>
omdurman- rules/src/ lib.rs:71 GH:omdurman- rules/src/lib.rs:71	GameTurnIndex::value	<pre> 69 70 impl \$name { 71 pub fn value(self) -> u16 { 72 match self { 73 \$(Self::\$variant => \$value,)+ </pre>
omdurman- rules/src/ effects.rs:792 GH:omdurman- rules/src/ effects.rs:792	PendingMelee	<pre> 790 #[serde(default)] 791 pub game_result: Option<crate::GameResult>, 792 } 793 794 /// A declared-but-unresolved melee attack, with its 795 pre-rolled dice held so </pre>

Covered by tests: `both_ready_auto_advances_out_of_setup` `fire_combat_wrong_phase_rejected` `new_game_starts_in_setup` `scenario_turn_dispatches_correctly` `turn_advances_through_phases` `game_turn_marker_cell_returns_none`

17/19 implemented

§5 – Movement Phase

§5 – Movement Phase (general)

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Movement Phase

Covered by tests: `disrupted_unit_cannot_fire` `disrupted_unit_may_not_act`

§5.3 – Constructing the Zariba

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Constructing the Zariba

The Zariba trench and thorn hedge hexsides are built and in place in the historical scenario only. These hexsides are considered clear terrain in the campaign game. The Anglo-Egyptian player may, however, find it useful to construct this defensive position during the campaign game. The Zariba hexsides may only be built in their position as displayed on the mapsheet. Construction procedure is as follows: any Anglo-Egyptian infantry unit that begins and ends the Anglo-Egyptian player turn adjacent to (and on the Nile side of) Zariba hexsides has constructed all Zariba hexsides to which he is adjacent. The constructing unit may neither fire offensively nor melee attack during the turn of construction. Use a blank counter to denote units constructing Zariba hexsides. See 9.23 for defensive benefits and movement restrictions of Zariba hexsides.

(see manual for full text)

See also: §9.23

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:822 GH:omdurman-rules/src/lib.rs:822	<code>constructing_zariba</code>	<pre>820 /// Set while the unit is building Zariba hexsides -- -- neither offensive 821 /// fire nor melee allowed that turn (§5.3). 822 pub constructing_zariba: bool, 823 /// Set when the Royal Engineers are committed to a demolition this turn 824 /// (§6.53) -- neither offensive fire nor melee allowed that turn.</pre>
omdurman-rules/src/effects.rs:114 GH:omdurman-rules/src/effects.rs:114	<code>ConstructZariba</code>	<pre>112 113 /// Begin constructing a Zariba hexside (rulebook §5.3). 114 ConstructZariba { 115 unit_ids: Vec<UnitId>, 116 hexside: HexsideRef,</pre>
omdurman-rules/src/effects.rs:2211 GH:omdurman-rules/src/effects.rs:2211	<code>apply_construct_zariba</code>	<pre>2209 apply_advance_after_combat(state, *unit_id, *to) 2210 } 2211 GameEffect::RecoverUnit { unit_id } => apply_recover_unit(state, *unit_id), 2212 GameEffect::ConstructZariba { unit_ids, hexside } => { 2213 apply_construct_zariba(state, unit_ids, *hexside)</pre>
omdurman-rules/src/lib.rs:841 GH:omdurman-rules/src/lib.rs:841	<code>UnitState::may_attack_this</code>	<pre>839 /// A unit that began construction this turn may not fire offensively or 840 /// melee (§5.3, §6.53). 841 pub fn may_attack_this_turn(self) -> bool { 842 !self.disrupted && !self.constructing_zariba</pre>

		<pre>&& !self.demolishing 843 }</pre>
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§5.11 – Movement allowances printed on units

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The movement allowances of the various unit types are printed directly on the units (see 2.3). A unit may move up to this printed movement allowance, paying varying costs for different terrain types (see the Terrain Effects Chart).

See also: §2.3

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:139 GH:omdurman-rules/src/lib.rs:139	MovementAllowance	<pre>137 /// is a named variant. 138 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 139 pub enum MovementAllowance { 140 /// Immobile (forts, wrecked gunboats). 141 Immobile = 0,</pre>
omdurman-rules/src/lib.rs:799 GH:omdurman-rules/src/lib.rs:799	UnitMovement	<pre>797 /// Movement allowance -- uniform for land units, split for gunboats (rulebook §5.11, §5.24, §5.25). 798 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 799 pub enum UnitMovement { 800 Land(MovementAllowance), 801 Gunboat(GunboatMovement),</pre>
omdurman-types/src/lib.rs:291 GH:omdurman-types/src/lib.rs:291	HexDirection	<pre>289 /// (`+q`, `+q+r`, `+r`, `-q`, `-q-r`, `-r` for pointy- top hexes) (rulebook §5.11, §5.24). 290 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, Default)] 291 pub enum HexDirection { 292 #[default] 293 East = 0,</pre>
omdurman-rules/src/lib.rs:181 GH:omdurman-rules/src/lib.rs:181	MovementPoints	<pre>179 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, Default, 180)] 181 pub struct MovementPoints(i16); 182 183 impl MovementPoints {</pre>
omdurman-rules/src/terrain_chart.rs:21 GH:omdurman-rules/src/terrain_chart.rs:21	terrain_effects_chart	<pre>19 /// 20 /// Source: printed Terrain Effects Chart on the mapsheet. 21 pub fn terrain_effects_chart(terrain: Terrain) -> TerrainEntry { 22 match terrain { 23 Terrain::Clear { .. } => TerrainEntry {</pre>
omdurman-rules/src/terrain_chart.rs:77 GH:omdurman-rules/src/terrain_chart.rs:77	movement_cost	<pre>75 /// Convenience: get the movement cost for a terrain type (rulebook §5.11, Terrain Effects Chart). 76 /// Returns `None` for impassable terrain (Nile). 77 pub fn movement_cost(terrain: Terrain) -> Option<MovementAllowance> { 78 terrain_effects_chart(terrain).movement_cost 79 }</pre>
omdurman-rules/src/terrain_chart.rs:85 GH:omdurman-rules/src/terrain_chart.rs:85	movement_cost_with_road	<pre>83 /// underlying terrain; without a road it's the terrain's own cost. The road is 84 /// a movement overlay only -- combat/LOS still use the underlying terrain. 85 pub fn movement_cost_with_road(terrain: Terrain, road: bool) -> Option<MovementAllowance> { 86 if road { 87 Some(MovementAllowance::One)</pre>

omdurman-types/src/lib.rs:422 GH:omdurman-types/src/lib.rs:422	Ter- rain::passable_by_land	<pre> 420 421 /// Whether this terrain may be entered by land units (rulebook \$5.11). 422 pub fn passable_by_land(self) -> bool { 423 !self.is_nile() 424 } </pre>
omdurman-types/src/lib.rs:488 GH:omdurman-types/src/lib.rs:488	Terrain::is_crossroad	<pre> 486 487 /// Whether roads converge at this hex's centre. 488 pub fn is_crossroad(self) -> bool { 489 matches!(self.road(), Road::Crossroad) 490 } </pre>
omdurman-hexmap/src/map.rs:11 GH:omdurman-hexmap/src/map.rs:11	GameMap::roads	<pre> 9 // -- Runtime game map ----- 10 11 /// Active hex map state: hex terrain, hexsides, roads, excluded hexes, and 12 /// the overlay parameters that define the grid shape and orientation. 13 #[derive(Resource, Default)] </pre>

Covered by tests: [clear_terrain_no_bonus](#) [nile_is_impassable](#) [rough_movement_and_defense](#) [swamp_movement_and_defense](#) [hilltop_movement_and_defense](#) [huts_movement_and_defense](#) [movement_cost_convenience_matches_chart](#) [movement_cost_with_road_always_one](#) [land_unit_may_not_enter_nile](#) [movement_cost_without_road_matches_terrain](#) [movement_cost_for_uses_terrain](#) [movement_cost_for_road_costs_one](#) [road_gives_crossroad](#)

§5.12 – Move up to allowance, hex by hex (cumulative MP cap)

[implemented](#)

manual page unknown

A player may move as many or as few of his units as desired during each movement phase, limited only by the units' movement allowance, the terrain costs paid in moving from hex to hex, and enemy zones of control (see 5.4).

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1897 GH:omdurman-rules/src/effects.rs:1897	mp_spent	<pre> 1895 self.units.iter().filter(u u.position == hex).collect() 1896 } 1897 1898 /// Movement points `unit_id` has already spent this turn (\$5.11/\$5.12). 1899 pub fn mp_spent(&self, unit_id: UnitId) -> i16 { </pre>

§5.13 – No MP accumulation between turns

[implemented](#)

manual page unknown

A unit may never accumulate movement points from turn to turn, nor may a unit transfer unused movement points to other units. A unit's unused movement points in any given turn are considered lost.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:2330 GH:omdurman-rules/src/effects.rs:2330	end_player_turn	<pre> 2328 Ok(()) 2329 } 2330 2331 /// End the current player's turn: recover disrupted units, switch active player, advance turn index (rulebook \$4). 2332 pub fn end_player_turn(state: &mut GameState) -> Result<(), RuleError> { </pre>

§5.21 – Friendlies transport via gunboat

implemented

manual page unknown

In general, naval transport missions are not allowed, i.e. gunboats may not carry any land units. The sole exception is that the Anglo-Egyptian player may transport the surviving units of the “Friendlies” brigade from the east bank of the Nile to the west bank after, and only after, the Dervish east bank unit (Isa Zachneih) has been eliminated. The transport is accomplished in the following sequence: a) on any turn that a “Friendlies” unit and any Anglo-Egyptian gunboat start their turn adjacent, that unit may load onto (i.e. stack with) the gunboat; b) during the Anglo-Egyptian player’s next turn the gunboat may move to any Nile hex adjacent to a west bank hex (up to the gunboat’s movement allowance); c) on the Anglo-Egyptian player’s third turn the “Friendlies” unit may disembark and move normally, paying the normal terrain cost for the first hex entered. The gunboat may also move normally that turn.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:647 GH:omdurman-rules/src/lib.rs:647	is_friendlys	<pre>645 /// "Friendlies" units obey several special rules (\$5.21, \$5.23, \$6.52, 646 /// \$9.14 victory conditions). 647 pub fn is_friendlys(&self) -> bool { 648 matches!(649 self,</pre>
omdurman-rules/src/lib.rs:819 GH:omdurman-rules/src/lib.rs:819	loaded_on	<pre>817 pub disrupted: bool, 818 /// `Some(gunboat)` after a "Friendlies" unit loads onto a gunboat (\$5.21). 819 pub loaded_on: Option<UnitId>, 820 /// Set while the unit is building Zariba hexsides -- neither offensive 821 /// fire nor melee allowed that turn (\$5.3).</pre>
omdurman-rules/src/effects.rs:141 GH:omdurman-rules/src/effects.rs:141	FriendliesTransport	<pre>139 140 /// Load/disembark the "Friendlies" brigade via gunboat (rulebook \$5.21). 141 FriendliesTransport(crate::FriendliesAction), 142 143 // -- Optional rules -----</pre>
omdurman-rules/src/effects.rs:4154 GH:omdurman-rules/src/effects.rs:4154	apply_friendlys_transport	<pre>4152 // the Dervish player rolls one die... made during the movement phase." 4153 if state.dervish_deserted { 4154 return Err(DesertionError::AlreadyDeserted.into()); 4155 } 4156 if state.scenario != Scenario::Campaign {</pre>
omdurman-rules/src/lib.rs:1081 GH:omdurman-rules/src/lib.rs:1081	FriendliesAction	<pre>1079 /// tracks each unit-gunboat pair independently. 1080 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1081 pub enum FriendliesAction { 1082 /// Turn N (the load turn): unit and gunboat started adjacent; unit 1083 /// loads onto (stacks with) the gunboat.</pre>
omdurman-rules/src/lib.rs:1097 GH:omdurman-rules/src/lib.rs:1097	TransportState	<pre>1095 /// third turn. 1096 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1097 pub enum TransportState { 1098 /// Turn N (the load turn): unit and gunboat started adjacent; unit 1099 /// loads onto (stacks with) the gunboat.</pre>

§5.22 – Land units may never enter a Nile River hex

implemented

manual page unknown

With the exception of 5.21, land units may never enter a Nile River hex. Only gunboats may enter and move along Nile River hexes.

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1282 GH:omdurman-rules/src/effects.rs:1282	<code>can_move_unit_to</code>	<pre>1280 /// ZOC may still move out (§5.43). 1281 /// 1282 /// The caller supplies `to` because the engine costs moves by distance and 1283 /// does not otherwise know the intervening hexes. The §5.44 hexside 1284 /// exceptions are applied by [`hex_in_enemy_zoc`] using the attached board.</pre>
omdurman-rules/src/effects.rs:1007 GH:omdurman-rules/src/effects.rs:1007	<code>in_deployment_zone</code>	<pre>1005 /// - **Historical / Campaign** (§9.211-9.212, §9.11): permissive. The 1006 /// manual's constraints there are the 13 Zariba hexes, the Kerreri huts, 1007 /// and per-leader "within three hexes" color groups -- data the engine's 1008 /// `BoardInfo` does not carry (no Zariba-hex set, no Kerreri landmark, no 1009 /// per-unit leader color), so those are enforced by the scenario set-up</pre>

Covered by tests: [fok_ae_gunboat_deploys_only_on_nile](#) [fok_ae_land_unit_rejected_on_nile](#) [deploy_via_real_sprite_resolution_matches_engine](#) [fok_dervish_land_unit_rejected_on_nile](#)

§5.23 – Walled city entry restrictions

implemented

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Only certain units may enter the walled portion of Omdurman. For the Dervish player these are the Khalifa unit, the three Dervish artillery units, and the Taiasha units (the Khalifa's bodyguard). Any Anglo-Egyptian units that can get to the walled city may enter it (except gunboats and "Friendlies"). Units entering and/or exiting the walled city may only do so through a gate or breach hexside.

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:141 GH:omdurman-types/src/lib.rs:141	<code>HexsideRef</code>	<pre>139 /// data by [`HexsideRef`]. 140 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 141 pub struct HexsideRef { 142 pub a: HexCoord, 143 pub b: HexCoord,</pre>
omdurman-types/src/lib.rs:176 GH:omdurman-types/src/lib.rs:176	<code>HexsideKind</code>	<pre>174 strum::EnumIter, 175)] 176 pub enum HexsideKind { 177 /// City wall (Khartoum, walled city of Omdurman). Blocks LOS, blocks 178 /// movement except across gates/breaches (§5.23), blocks ZOC into the city</pre>
omdurman-types/src/lib.rs:247 GH:omdurman-types/src/lib.rs:247	<code>blocks_movement</code>	<pre>245 /// `omdurman-rules`. The trench *end* variants are therefore intentionally 246 /// not blocking. 247 pub fn blocks_movement(self) -> bool {</pre>

		<pre> 248 matches!(249 self, </pre>
omdurman-rules/src/lib.rs:673 GH:omdurman-rules/src/lib.rs:673	may_enter_walled_city	<pre> 671 /// Taiasha bodyguard may enter. Anglo-Egyptian: any unit that can reach the 672 /// walled city *except* gunboats and "Friendlies". 673 pub fn may_enter_walled_city(&self) -> bool { 674 match self { 675 // §5.23 Dervish: Khalifa, artillery, Taiasha. </pre>
omdurman-rules/src/board.rs:228 GH:omdurman-rules/src/board.rs:228	is_walled_city	<pre> 226 /// touches it on one. The two-sided threshold keeps the predicate robust to 227 /// a map edit that adds or removes a single wall segment. 228 pub fn is_walled_city(&self, hex: HexCoord) -> bool { 229 if matches!(230 self.location_at(hex), </pre>
omdurman-rules/src/effects.rs:304 GH:omdurman-rules/src/effects.rs:304	WalledCityEntry	<pre> 302 303 #[error("unit {0:?} is not eligible to enter the walled city of Omdurman at {1:?} (§5.23)")] 304 WalledCityEntry(UnitId, HexCoord), 305 306 #[error("movement cost {cost:?} exceeds allowance {allowance:?}")] </pre>

Covered by tests: [can_move_rejects_wall_hexside](#) [can_move_allows_gate_hexside](#) [walled_city_entry_allows_khalifa](#) [walled_city_entry_rejects_unauthorized_dervish](#) [walled_city_entry_rejects_ae_gunboat](#) [walled_city_entry_not_enforced_for_fok](#)

§5.24 – Gunboat upstream/downstream movement

[implemented](#)

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Note that gunboats have two movement allowances separated by a slash, e.g. 10/16. The smaller number is the movement allowance when moving upstream, i.e. against the current (the direction of the current is indicated by arrows in the Nile). The larger number is the movement allowance when moving downstream, i.e. with the current. Gunboats may combine movement in both directions, but if they move even one hex upstream, their upstream movement allowance is their maximum movement allowance for that turn, and may not be exceeded.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:581 GH:omdurman-rules/src/lib.rs:581	GunboatMovement	<pre> 579 /// the turn. 580 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 581 pub struct GunboatMovement { 582 pub upstream: MovementAllowance, 583 pub downstream: MovementAllowance, </pre>
omdurman-types/src/lib.rs:864 GH:omdurman-types/src/lib.rs:864	is_boat	<pre> 862 863 /// Gunboats use the split upstream/downstream movement allowance (§5.24). 864 pub fn is_boat(self) -> bool { 865 matches!(self, UnitKind::Gunboat { .. }) 866 } </pre>

Covered by tests: [boat_annotation_yields_split_gunboat_movement](#)

§5.25 – Dervish forts may not move

implemented

manual page unknown

Dervish forts may not move in any way once placed.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:803 GH:omdurman-rules/src/lib.rs:803	Immobile	<pre>801 Gunboat(GunboatMovement), 802 /// Forts may not move once placed (§5.25). 803 Immobile, 804 } 805 </pre>
omdurman-types/src/lib.rs:825 GH:omdurman-types/src/lib.rs:825	UnitKind::Fort	<pre>823 /// Permanent emplacement (§6.54): fire (artillery) / melee (defensive). 824 /// May not move once placed (§5.25). 825 Fort { fire: i32, melee: i32 }, 826 /// Dervish leader (§6.51): fire / melee / movement. May melee attack (§7.4). 827 DervishLeader { fire: i32, melee: i32, movement: 828 i32 },</pre>
omdurman-rules/src/lib.rs:803 GH:omdurman-rules/src/lib.rs:803	UnitMovement::Immobile	<pre>801 Gunboat(GunboatMovement), 802 /// Forts may not move once placed (§5.25). 803 Immobile, 804 } 805 </pre>

Covered by tests: [forts_are_never_advance_eligible](#)

§5.26 – Units stop on entering enemy ZOC

implemented

manual page unknown

Units must stop their movement immediately upon entering an enemy zone of control (see 5.4).

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1282 GH:omdurman-rules/src/effects.rs:1282	can_move_unit_to	<pre>1280 /// ZOC may still move out (§5.43). 1281 /// 1282 /// The caller supplies `to` because the engine costs moves by distance and 1283 /// does not otherwise know the intervening hexes. The §5.44 hexside 1284 /// exceptions are applied by [<code>hex_in_enemy_zoc</code>] using the attached board.</pre>
omdurman-rules/src/effects.rs:2068 GH:omdurman-rules/src/effects.rs:2068	hex_in_enemy_zoc	<pre>2066 /// Applies the §5.44 hexside exceptions using the attached board: a ZOC does 2067 /// not extend across a khor/wall/Zariba hexside, and (except for gunboats) 2068 /// does not extend into or out of a Nile hex. With no board loaded these 2069 /// reduce to the plain adjacency rule. 2070 pub fn hex_in_enemy_zoc{</pre>

§5.41 – All units except AE leaders exert ZOC

implemented

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All units except Anglo-Egyptian leaders exert a zone of control (hereafter called a ZOC) into their six adjacent hexes (exception: Gunboats exert a ZOC only against enemy gunboats). Disrupted units have no ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:871 GH:omdurman-rules/src/lib.rs:871	ZocReason	<pre> 869 /// Used by the engine when answering "is this hex in an enemy ZOC?". 870 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 871 pub enum ZocReason { 872 /// Normal ZOC: any non-disrupted unit other than an Anglo-Egyptian 873 /// leader (§5.41) projects ZOC into each of its six adjacent hexes. </pre>
omdurman-rules/src/effects.rs:2033 GH:omdurman-rules/src/effects.rs:2033	unit_projects_zoc	<pre> 2031 /// Returns the [<code>`ZocReason`</code>] when ZOC applies, else <code>`None`</code>. The hexside 2032 /// subtleties (walls/gates/khor/forts/Zariba block or redirect ZOC -- 2033 /// §5.44) need the game map, which the engine does not hold; the app layers 2034 /// those on top. This is the position/kind/ disruption core of the rule. 2035 pub fn unit_projects_zoc(</pre>

§5.42 – No MP cost to enter/leave enemy ZOC

implicit

manual page unknown

There is no movement point cost to enter or leave an enemy ZOC.

Covered by tests: [entering_enemy_zoc_costs_no_extra_mp](#)

§5.43 – Units stop when entering enemy ZOC

implemented

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All units must stop when they enter an enemy ZOC and may move no further that turn. In their next movement phase they may withdraw or, if desired, move directly into another enemy ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1282 GH:omdurman-rules/src/effects.rs:1282	can_move_unit_to	<pre> 1280 /// ZOC may still move out (§5.43). 1281 /// 1282 /// The caller supplies <code>`to`</code> because the engine costs moves by distance and 1283 /// does not otherwise know the intervening hexes. The §5.44 hexside 1284 /// exceptions are applied by [<code>`hex_in_enemy_zoc`</code>] using the attached board. </pre>
omdurman-rules/src/effects.rs:2068 GH:omdurman-rules/src/effects.rs:2068	hex_in_enemy_zoc	<pre> 2066 /// Applies the §5.44 hexside exceptions using the attached board: a ZOC does 2067 /// not extend across a khor/wall/Zariba hexside, and (except for gunboats) 2068 /// does not extend into or out of a Nile hex. With no board loaded these 2069 /// reduce to the plain adjacency rule. 2070 pub fn hex_in_enemy_zoc(</pre>

§5.44 – ZOC limitations (walls, khor, fort, Nile, Zariba)

implemented

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ZOCs do not extend into or out of a Nile River hex (exception: Gunboats, see 5.41). ZOCs do not extend across a khor, into a fort, or into a hex inside the walled city across a wall hexside. ZOCs do extend out of a fort (even if unoccupied), and from a walled city hex into an adjacent non-walled-city hex across a wall hexside. ZOCs also extend out of (but not into) a walled city hex across a gate hexside. ZOCs extend both ways across a breach hexside. ZOCs also extend out of, but not into, a hut or building hex. In the historical scenario ZOCs extend out of, but not into, the Zariba across a Zariba hexside (also in the campaign game if the Zariba is constructed).

(see manual for full text)

See also: §5.41

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:871 GH:omdurman-rules/src/lib.rs:871	ZocReason	<pre>869 /// Used by the engine when answering "is this hex in 870 an enemy ZOC?". 871 #[derive(Serialize, Deserialize, Clone, Copy, 872 PartialEq, Eq, Debug)] 873 pub enum ZocReason { 874 /// Normal ZOC: any non-disrupted unit other than 875 an Anglo-Egyptian 876 /// leader (§5.41) projects ZOC into each of its 877 six adjacent hexes.</pre>
omdurman-types/src/lib.rs:181 GH:omdurman-types/src/lib.rs:181	Wall	<pre>179 /// (§5.44), blocks melee (§7.2), blocks advance- 180 after-combat (§6.82). 181 #[default] 182 Wall, 183 /// Gate hexside in a wall. ZOC extends *out of* 184 the walled city through 185 /// gates but not into it (§5.44). Melee may be 186 made through a gate (§7.2).</pre>
omdurman-types/src/lib.rs:190 GH:omdurman-types/src/lib.rs:190	Khor	<pre>188 /// Khor -- gully/wadi. ZOCs do not extend across 189 (§5.44); advance after 190 /// combat may not cross (§6.82). 191 Khor, 192 /// Crest line. Blocks LOS unless the firer is on 193 the higher side 194 /// (§6.3 note 7).</pre>
omdurman-rules/src/lib.rs:884 GH:omdurman-rules/src/lib.rs:884	ZocReason::Zariba	<pre>882 /// Zariba hexside ZOC behaviour in the historical 883 scenario / when the 884 /// Zariba is constructed (§5.44). 885 Zariba, 886 }</pre>
omdurman-rules/src/effects.rs:2033 GH:omdurman-rules/src/effects.rs:2033	unit_projects_zoc	<pre>2031 /// Returns the [<code>ZocReason</code>] when ZOC applies, 2032 else <code>None</code>. The hexside 2033 /// subtleties (walls/gates/khor/forts/Zariba 2034 block or redirect ZOC -- 2035 /// §5.44) need the game map, which the engine 2036 does not hold; the app layers 2037 /// those on top. This is the position/kind/ 2038 disruption core of the rule. 2039 pub fn unit_projects_zoc(</pre>
omdurman-types/src/lib.rs:264	HexsideKind::blocks_zoc	<pre>262 /// cannot express; those are left to the caller. 263 This predicate captures the 264 /// symmetric "does not extend across" cases.</pre>

GH:omdurman-types/src/lib.rs:264		<pre> 264 pub fn blocks_zoc(self) -> bool { 265 matches!(266 self, </pre>
omdurman-rules/src/effects.rs:2068 GH:omdurman-rules/src/effects.rs:2068	hex_in_enemy_zoc	<pre> 2066 /// Applies the §5.44 hexside exceptions using the attached board: a ZOC does 2067 /// not extend across a khor/wall/Zariba hexside, and (except for gunboats) 2068 /// does not extend into or out of a Nile hex. With no board loaded these 2069 /// reduce to the plain adjacency rule. 2070 pub fn hex_in_enemy_zoc(</pre>

§5.51 – Stacking limit (4 units + leaders, gunboats isolated)

implemented

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No more than four units may occupy a hex, with the exception of leaders and gunboats. All leader units are free stacking, i.e. they may stack in addition to the four-unit-per-hex stacking limitation. Gunboats may not stack with any other unit (Exception: 5.21). Players may move through friendly units at no additional cost in movement points. The stacking limitation applies only at the end of the movement phase and during combat.

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:893 GH:omdurman-rules/src/lib.rs:893	OverLimit	<pre> 891 /// and the gunboat exception. 892 #[error("hex stack exceeds the four-unit limit")] 893 OverLimit, 894 /// "Gunboats may not stack with any other unit" (§5.51, exception §5.21). 895 #[error("gunboats may not stack with non-gunboat units")] </pre>
omdurman-rules/src/lib.rs:896 GH:omdurman-rules/src/lib.rs:896	GunboatStack	<pre> 894 /// "Gunboats may not stack with any other unit" (§5.51, exception §5.21). 895 #[error("gunboats may not stack with non-gunboat units")] 896 GunboatStack, 897 /// "Units of different Dervish tribes may not stack together" (§5.52). 898 #[error("Dervish units of different tribes may not stack")] </pre>
omdurman-rules/src/effects.rs:1953 GH:omdurman-rules/src/effects.rs:1953	check_stacking	<pre> 1951 /// * §5.51 -- at most four units per hex, *excluding* free-stacking leaders 1952 /// and gunboats; gunboats may not share a hex with any other unit. 1953 /// * §5.52 -- units of different Dervish tribes may not stack together. 1954 /// * §5.53 -- a Dervish leader may stack only with units of its command. 1955 pub fn check_stacking(</pre>

Covered by tests: `stacking_over_limit_rejected` `mid_move_stacking_allows_pass_through` `mid_move_stacking_rejects_over_limit_destination`

§5.52 – Different Dervish tribes may not stack together

implemented

manual page unknown

The units of different Dervish tribes may not stack together, even if they are the same color (e.g. although both are green, Mulazmin and Jehadia units may not stack with each other).

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:899 GH:omdurman-rules/src/lib.rs:899	DervishTribeMix	<pre> 897 /// "Units of different Dervish tribes may not stack together" (§5.52). 898 #[error("Dervish units of different tribes may not stack")] 899 DervishTribeMix, 900 /// "If Dervish leaders elect to stack, they may only stack with units of 901 /// their command (i.e. colour)" (§5.53).</pre>

Covered by tests: [green_sections_are_mulazmin_tribal_units](#) [deploy_rejects_dervish_tribe_mix](#)

§5.53 – Leader stacking with command colour only

implemented

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Leader units are not required to stack. If Dervish leaders elect to stack, however, they may only stack with units of their command (i.e. color). For example, Sheik El Din may only stack with Mulazmins or Jehadias.

****5.54) Anglo-Egyptian Brigade Integrity:**** All British, Sudanese, and Egyptian infantry units have their brigade designation printed in the upper right corner (e.g. “2B” = 2nd British Brigade; “3E” = 3rd Egyptian Brigade, etc.). In any combat phase in which all four infantry battalions belonging to any Anglo-Egyptian infantry brigade are stacked in the same hex they are said to have brigade integrity. Stacks having brigade integrity receive a +1 modifier to their fire combat die roll provided they all fire at the same enemy occupied hex. This modifier is in addition to the normal +1 bonus given to all Anglo-Egyptian direct fire attacks (see 6.24).

(see manual for full text)

See also: §5.54, §6.24

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:903 GH:omdurman-rules/src/lib.rs:903	DervishLeaderCommandMismatch	<pre> 901 /// their command (i.e. colour)" (§5.53). 902 #[error("Dervish leader may only stack with units of their own command")] 903 DervishLeaderCommandMismatch, 904 } 905 </pre>

§5.54 – Anglo-Egyptian Brigade Integrity

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:761 GH:omdurman-rules/src/lib.rs:761	BrigadeIntegrity	<pre> 759 /// Only a full stack of four battalions qualifies. Three or fewer may still 760 /// stack and fire, but they receive no brigade- integrity bonus. 761 pub fn brigade_integrity(identities: &[UnitIdentity]) - > BrigadeIntegrity { 762 let Some(brigade) = identities.first().and_then(i i.brigade()) else { 763 return BrigadeIntegrity::None;</pre>
omdurman-rules/src/lib.rs:761 GH:omdurman-rules/src/lib.rs:761	brigade_integrity	<pre> 759 /// Only a full stack of four battalions qualifies. Three or fewer may still 760 /// stack and fire, but they receive no brigade- integrity bonus. 761 pub fn brigade_integrity(identities: &[UnitIdentity]) - > BrigadeIntegrity {</pre>

		<pre> 762 let Some(brigade) = identities.first().and_then(i i.brigade()) else { 763 return BrigadeIntegrity::None; </pre>
omdurman-rules/src/lib.rs:761 GH:omdurman-rules/src/lib.rs:761	FireModifier::BrigadeIntegrity	<pre> 759 /// Only a full stack of four battalions qualifies. Three or fewer may still 760 /// stack and fire, but they receive no brigade- integrity bonus. 761 pub fn brigade_integrity(identities: &[UnitIdentity]) - > BrigadeIntegrity { 762 let Some(brigade) = identities.first().and_then(i i.brigade()) else { 763 return BrigadeIntegrity::None; </pre>
omdurman-rules/src/lib.rs:368 GH:omdurman-rules/src/lib.rs:368	BattalionOrdinal	<pre> 366 /// brigade integrity requires all four stacked in one hex (\$5.54). 367 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 368 pub enum BattalionOrdinal { 369 First = 1, 370 Second = 2, </pre>
omdurman-types/src/lib.rs:923 GH:omdurman-types/src/lib.rs:923	BrigadeId	<pre> 921 /// same field for uniform handling. 922 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 923 pub struct BrigadeId { 924 pub number: u8, 925 pub nationality: BrigadeNationality, </pre>

Covered by tests:

- [brigade_designation_ignored_for_non_infantry](#)
- [printed_brigade_designation_overrides_column](#)
- [tribe_stats_come_from_annotation](#)
- [section_owner_anglo_egyptian_sections](#)
- [section_owner_green_sections_are_dervish](#)
- [brigade_integrity_four_battalions_returns_integrated](#)
- [ae_infantry_fourth_battalion_from_col_3](#)
- [brigade_integrity_empty_slice](#)
- [brigade_integrity_friendlys_returns_none](#)
- [section_owner_dervish_sections](#)
- [brigade_integrity_three_battalions_returns_none](#)
- [unit_identity_brigade_and_battalion_accessors](#)
- [ae_infantry_brigade_number_three_from_col_7](#)
- [brigade_integrity_non_infantry_returns_none](#)
- [brigade_integrity_mixed_brigades_returns_none](#)
- [ae_infantry_third_battalion_from_col_2](#)

21/25 implemented

§6 – Fire Combat Phase

§6.3 – Line of Sight Table

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Line of Sight Table

This table is located on the back of this rulebook and should be self-explanatory. Locate the terrain type the firing unit is in and cross-index it with the terrain type the target unit is in. Terrain types in the intersecting box block line of sight, with exceptions as footnoted. Also study the “Special LOS Notes” given and remember that (with the exception of howitzer fire — see 6.64) you can’t fire at anything you can’t see!

See also: §6.64

File	Symbol	Code Snippet
omdurman-rules/src/los_table.rs:55 GH:omdurman-rules/src/los_table.rs:55	LosLevel	<pre>53 /// Ordered lowest to highest: `Ground < Rough < 54   Hilltop`. 55 #[derive(Clone, Copy, PartialEq, Eq, PartialOrd, Ord, 56 Hash, Debug)] 57 pub enum LosLevel { 58 Ground, 59 Rough,</pre>
omdurman-rules/src/los_table.rs:63 GH:omdurman-rules/src/los_table.rs:63	LosFeature	<pre>61 /// A feature on the LOS ray that may block (rulebook 62 §6.3). 63 #[derive(Clone, Copy, PartialEq, Eq, Hash, Debug)] 64 pub enum LosFeature { 65 /// A hex containing units (gunboats/forts excluded 66 per note a). 67 Units,</pre>
omdurman-rules/src/los_table.rs:83 GH:omdurman-rules/src/los_table.rs:83	LosCondition	<pre>81 /// A positional condition from the LOS table Detail 82 footnotes. 83 #[derive(Clone, Copy, PartialEq, Eq, Debug)] 84 pub enum LosCondition { 85 /// (1) Blocks only if the ray passes through more 86 than two such features. 87 MoreThanTwo,</pre>
omdurman-rules/src/los_table.rs:121 GH:omdurman-rules/src/los_table.rs:121	los_level	<pre>119 120 /// Map a terrain type to its LOS level (rulebook 121 §6.3). 122 pub fn los_level(terrain: Terrain) -> LosLevel { 123 use LosLevel::*; 124 match terrain {</pre>
omdurman-rules/src/los_table.rs:139 GH:omdurman-rules/src/los_table.rs:139	los_level_for_unit	<pre>137 /// 138 /// For all other units, the level is derived from the 139 terrain at `hex`. 140 pub fn los_level_for_unit(141 kind: UnitKind, 142 hex: HexCoord,</pre>
omdurman-rules/src/los_table.rs:175 GH:omdurman-rules/src/los_table.rs:175	blocking_rules	<pre>173 /// if ALL conditions are satisfied (AND semantics). An 174 empty conditions slice 175 /// means the feature always blocks. 176 pub fn blocking_rules(177 firer: LosLevel, 178 target: LosLevel,</pre>
omdurman-rules/src/los_table.rs:330 GH:omdurman-	has_los	<pre>328 /// `unit_level_at` closure returns the LOS level of 329 blocking units 330 /// (non-gunboat, non-fort per note a) in an 331 intervening hex, or `None`. 332 pub fn has_los(</pre>

rules/src/ los_table.rs:330		331 board: &crate::board::BoardInfo, 332 from: HexCoord,
omdurman- types/src/ lib.rs:219 GH:omdurman- types/src/lib.rs:219	HexsideKind::blocks_los	217 /// (LOS table conditions 2-4, 7) and note (e) are handled by the engine 218 /// in `omdurman_rules::los_table`, not by this predicate. 219 pub fn blocks_los(self) -> bool { 220 matches!(self, HexsideKind::Wall HexsideKind::Crest) 221 }
omdurman- types/src/ lib.rs:219 GH:omdurman- types/src/lib.rs:219	Terrain::blocks_los	217 /// (LOS table conditions 2-4, 7) and note (e) are handled by the engine 218 /// in `omdurman_rules::los_table`, not by this predicate. 219 pub fn blocks_los(self) -> bool { 220 matches!(self, HexsideKind::Wall HexsideKind::Crest) 221 }
omdurman- types/src/ lib.rs:439 GH:omdurman- types/src/lib.rs:439	Terrain::is_los_trees	437 /// (\$6.3 note 1). Retained for compatibility; the full LOS engine 438 /// checks `Terrain::Trees` directly. 439 pub fn is_los_trees(self) -> bool { 440 matches!(self, Terrain::Trees { .. }) 441 }

Covered by tests: los_level_mapping blocking_rules_all_cells_covered has_los_empty_board_is_clear
has_los_adjacent_clear has_los_howitzer_bypasses has_los_wall_hexside_blocks_ground_to_ground has_los_gate_hexside_passes
has_los_breach_hexside_passes has_los_rough_intervening_blocks_ground_to_ground has_los_two_tree_hexes_pass_ground_to_ground
has_los_three_tree_hexes_block_ground_to_ground
has_los_two_hut_hexes_pass_ground_to_ground has_los_three_hut_hexes_block_ground_to_ground has_los_hilltop_to_hilltop_clear_no_units
has_los_hilltop_to_hilltop_blocked_by_hilltop_unit has_los_hilltop_to_hilltop_not_blocked_by_ground_unit
has_los_rough_to_rough_unit_at_lower_level_passes has_los_rough_to_rough_unit_at_same_level_blocks
has_los_rough_to_rough_hilltop_blocks has_los_ground_to_hilltop_intervening_hilltop_blocks
has_los_building_blocks_like_huts_ground_to_ground
has_los_two_building_hexes_pass_ground_to_ground los_level_for_unit_gunboat_is_rough los_level_for_unit_fort_is_ground
los_level_for_unit_walled_city_adj_wall_is_rough gunboat_firer_uses_rough_row_not_ground

§6.6 – Special Artillery Capabilities

implemented

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Special Artillery Capabilities

File	Symbol	Code Snippet
omdurman- rules/src/ lib.rs:513 GH:omdurman- rules/src/lib.rs:513	WeaponClass	511 /// enum so a "spear" unit cannot accidentally fire on the "Howitzer" line. 512 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 513 pub enum WeaponClass { 514 /// Dervish spears and swords -- no ranged fire at all. 515 Melee,

§6.7 – Defensive Fire

implemented

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Defensive Fire

In Defensive Fire phase, all of the non-moving player's units may fire at any of the moving player's units in range, within the limitations imposed by the rules of combat (see 6.1 to 6.6). There is no advance after combat as a result of defensive fires.

See also: §6.6

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:3374 GH:omdurman-rules/src/effects.rs:3374	<code>can_advance_after_combat</code>	<pre>3372 } 3373 // Retreat is a *reaction* to a declared *infantry* melee attack on the 3374 // unit's hex (§7.5): there must be a pending melee targeting where it 3375 // stands, made by at least one infantry attacker. 3376 match &self.pending_melee {</pre>

Covered by tests: `no_advance_after_defensive_fire` `defensive_fire_opens_no_advance_window`

§6.11 – Fire combat factor printed on units

implemented

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The fire combat factor of the various unit types is printed directly on the units and is a numerical expression of the unit's fire strength.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:94 GH:omdurman-rules/src/lib.rs:94	<code>FireFactor</code>	<pre>92 /// Every possible value from the annotated counter set is a named variant. 93 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 94 pub enum FireFactor { 95 One = 1, 96 Three = 3,</pre>

Covered by tests: `fire_factor_sum_to_row`

§6.12 – Fire combat is always voluntary

implicit

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Fire combat is always voluntary. A unit is never required to fire at enemy units merely because they are in range or adjacent.

§6.13 – Fire factor is unitary (may not be divided)

implicit

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If a unit elects to fire, its fire combat factor at an enemy unit, that fire combat factor is unitary. A unit's fire combat factor may not be divided up to fire at enemy units on different hexes.

§6.14 – Players may combine fire factors into one attack

implemented

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Players may combine fire during fire combat phase, i.e. they may fire at an enemy-occupied hex with as many friendly units as may legally do so, combining all of their fire combat factors into one attack. Note that in any given fire combat phase, however, a combat unit may only fire once and may only be fired at once (exceptions: Maxim guns and gunboats — see 6.4).

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:108 GH:omdurman-rules/src/lib.rs:108	<code>sum_to_row</code>	<pre> 106 impl FireFactor { 107 /// Sum multiple fire factors and return the corresponding Combat Results Table row (rulebook §6.11). 108 pub fn sum_to_row<'a>(factors: impl IntoIterator<Item = &'a FireFactor>) -> FireFactorRow { 109 let total: u16 = factors.into_iter().map(f f.value()).sum(); 110 crate::combat_results_table::FireFactorRow::from_total(total) </pre>

Covered by tests: `unit_may_only_be_fired_at_once_per_phase` `gunboat_and_maxim_may_be_fired_at_repeatedly`

§6.15 – May divide a stack to fire at different hexes

`implicit`

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Players may also divide a stack of units in order to fire at different enemy-occupied hexes. Anglo-Egyptian infantry units having brigade integrity, however, do not receive their +1 direct fire modifier unless they all fire at the same enemy-occupied hex (see 5.54).

See also: §5.54

§6.16 – Halving fire strength rounds down, minimum 1

`implemented`

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When halving fire combat strength, always round down each individual unit. For example, an Egyptian brigade of four battalions, each having a printed strength of 9 fire factors, will fire a total of 16 factors when halved. However, a unit's firing strength is never reduced below one by halving.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:548 GH:omdurman-rules/src/lib.rs:548	<code>RangeBand</code>	<pre> 546 /// multiplied at a given distance (§6.22). 547 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 548 pub enum RangeBand { 549 Tripled, 550 Doubled, </pre>

§6.21 – First check LOS before firing

`implemented`

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When combat units wish to fire at enemy units, first check the Line of Sight Table to be sure the firing unit can see the target hex (exception: howitzer fire, see 6.64).

See also: §6.64

File	Symbol	Code Snippet
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omdurman-rules/src/los_table.rs:175 GH:omdurman-rules/src/los_table.rs:175	blocking_rules	173 /// if ALL conditions are satisfied (AND semantics). An empty conditions slice 174 /// means the feature always blocks. 175 pub fn blocking_rules(176 firer: LosLevel, 177 target: LosLevel,
omdurman-rules/src/los_table.rs:330 GH:omdurman-rules/src/los_table.rs:330	has_los	328 /// `unit_level_at` closure returns the LOS level of blocking units 329 /// (non-gunboat, non-fort per note a) in an intervening hex, or `None`. 330 pub fn has_los(331 board: &crate::board::BoardInfo, 332 from: HexCoord,

Covered by tests: [can_fire_at_rejects_blocked_los](#) [can_fire_at_allows_clear_los](#)

§6.22 – Consult Range Effects Table

[implemented](#)

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Next consult the Range Effects Table to see if the firing unit's fire combat factor is tripled, doubled, normal, halved, or if the target hex is out of range. Add up the total number of fire combat factors firing at the enemy-occupied hex.

File	Symbol	Code Snippet
omdurman-rules/src/range_effects.rs:23 GH:omdurman-rules/src/range_effects.rs:23	ae_range_effects	21 /// Look up the range band for an Anglo-Egyptian weapon (§6.22, §6.24). 22 /// Distances > 10 are out of range for all weapons. 23 pub fn ae_range_effects(weapon: WeaponClass, distance: HexDistance) -> RangeBand { 24 if distance.value() > 10 { 25 return RangeBand::OutOfRange;
omdurman-rules/src/range_effects.rs:61 GH:omdurman-rules/src/range_effects.rs:61	dervish_range_effects	59 /// Look up the range band for a Dervish weapon (§6.22). 60 /// Distances > 10 are out of range for all weapons. 61 pub fn dervish_range_effects(weapon: WeaponClass, distance: HexDistance) -> RangeBand { 62 if distance.value() > 10 { 63 return RangeBand::OutOfRange;
omdurman-rules/src/lib.rs:532 GH:omdurman-rules/src/lib.rs:532	Range	530 /// (rulebook §6.22). Distances beyond 10 hexes are out of range for all weapons. 531 #[derive(Clone, Copy, PartialEq, Eq, Debug)] 532 pub enum Range { 533 One, 534 Two,
omdurman-rules/src/lib.rs:548 GH:omdurman-rules/src/lib.rs:548	RangeBand	546 /// multiplied at a given distance (§6.22). 547 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 548 pub enum RangeBand { 549 Tripled, 550 Doubled,
omdurman-rules/src/lib.rs:195 GH:omdurman-rules/src/lib.rs:195	HexDistance	193 /// (rulebook §6.22, §7.5). 194 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 195 pub struct HexDistance(u16); 196 197 impl HexDistance {

Covered by tests: [ae_rifles_doubled_at_range_1](#) [ae_rifles_halved_at_range_4](#) [ae_howitzer_range](#) [dervish_rifles_shorter_range](#) [melee_only_range_1](#) [ae_range_effects_artillery_full](#) [ae_range_effects_maxims_match_rifles](#) [ae_range_effects_distance_over_10](#) [ae_range_effects_howitzer_halved_4_to_10](#) [dervish_range_effects_rifles](#) [dervish_range_effects_artillery](#) [dervish_range_effects_maxims_and_howitzer](#)

dervish_range_effects_melee dervish_range_effects_distance_over_10 fire_combat_eliminate_target max_day_range_all_combos
range_effects_every_cell_ae_artillery range_effects_every_cell_ae_howitzer range_effects_every_cell_ae_maxims range_effects_every_cell_ae_rifles
range_effects_every_cell_dervish_artillery range_effects_every_cell_dervish_maxims_howitzer range_effects_every_cell_dervish_rifles
range_effects_every_cell_dervish_spears can_fire_at_gates_phase_range_and_player

§6.23 – Terrain defensive modifier

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Next check the Terrain Effects Chart to see if the enemy-occupied hex fired upon contains any terrain which gives the enemy units in that hex a defensive benefit. If so, apply this negative modifier to the roll of the ten-sided die and cross-index your net die roll on the Combat Results Table with the number of combat factors firing.

File	Symbol	Code Snippet
omdurman-rules/src/terrain_chart.rs:59 GH:omdurman-rules/src/terrain_chart.rs:59	defense_modifier	<pre> 57 58 /// Convenience: get the defense modifier for a terrain type (rulebook §6.23, Terrain Effects Chart). 59 pub fn defense_modifier(terrain: Terrain) -> i16 { 60 terrain_effects_chart(terrain).defense_modifier 61 } </pre>
omdurman-rules/src/lib.rs:930 GH:omdurman-rules/src/lib.rs:930	FireModifier::Terrain	<pre> 928 /// Negative modifier from the Terrain Effects Chart applied to the 929 /// defender's hex (§6.23). 930 Terrain(i16), 931 /// -2 thorn-hedge defensive modifier (§9.231). 932 ZaribaThornHedge, </pre>

Covered by tests: clear_terrain_no_bonus building_gives_minus_3 palm_grove_gives_minus_2 rough_movement_and_defense swamp_movement_and_defense hilltop_movement_and_defense huts_movement_and_defense defense_modifier_convenience_matches_chart

§6.24 – Anglo-Egyptian direct fire accuracy bonus and brigade integrity

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All Anglo-Egyptian direct fire attacks receive a +1 modifier to their die roll as an accuracy bonus. In addition, any stack of Anglo-Egyptian infantry having brigade integrity (see 5.54) receives a +1 modifier to their die roll if all four fire at the same enemy-occupied hex. These modifiers are cumulative.

See also: §5.54

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:924 GH:omdurman-rules/src/lib.rs:924	AngloEgyptianDirectFire	<pre> 922 pub enum FireModifier { 923 /// +1 to all Anglo-Egyptian *direct* fire (§6.24). 924 AngloEgyptianDirectFire, 925 /// +1 brigade integrity, applied only if all four battalions fire at 926 /// the same enemy-occupied hex (§5.54, §6.24). </pre>
omdurman-rules/src/lib.rs:761 GH:omdurman-rules/src/lib.rs:761	BrigadeIntegrity	<pre> 759 /// Only a full stack of four battalions qualifies. Three or fewer may still 760 /// stack and fire, but they receive no brigade- integrity bonus. 761 pub fn brigade_integrity(identities: &[UnitIdentity]) - > BrigadeIntegrity { 762 let Some(brigade) = identities.first().and_then(i i.brigade()) else { 763 return BrigadeIntegrity::None; </pre>
omdurman-rules/src/	FireModifier::die_modifier	<pre> 938 impl FireModifier { 939 /// Return the numeric die-roll modifier for this </pre>

lib.rs:940 GH:omdurman-rules/src/lib.rs:940		bonus/penalty (rulebook §6.24, §5.54, §6.23, §9.231, §9.232). 940 pub fn die_modifier(self) -> i16 { 941 match self { 942 FireModifier::AngloEgyptianDirectFire FireModifier::BrigadeIntegrity => 1,
--	--	--

§6.41 – Direct Fire Subphase

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:343 GH:omdurman-rules/src/lib.rs:343	DirectFire	341 pub enum FireSubPhase { 342 /// Direct fire (§6.41). Both sides participate in this sub-phase. 343 DirectFire, 344 /// Anglo-Egyptian only: Maxim second fire + named- gunboat howitzer fire (§6.42). 345 MaximSecondAndHowitzer,

§6.42 – Maxim Second Fire and Howitzer Fire Subphase

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:345 GH:omdurman-rules/src/lib.rs:345	MaximSecondAndHowitzer	343 DirectFire, 344 /// Anglo-Egyptian only: Maxim second fire + named- gunboat howitzer fire (§6.42). 345 MaximSecondAndHowitzer, 346 } 347
omdurman-types/src/lib.rs:876 GH:omdurman-types/src/lib.rs:876	fires_twice	874 /// Maxim guns fire twice per turn -- once in the Direct Fire Subphase and 875 /// again in the Maxim Second Fire Subphase (rulebook §6.42). 876 pub fn fires_twice(self) -> bool { 877 matches!(self, UnitKind::Maxim { .. }) 878 }

Covered by tests: [howitzer_on_target_7_to_10](#) [howitzer_scatters_below_7](#) [howitzer_short_on_5_6](#) [howitzer_long_on_3_4](#) [howitzer_left_right_on_1_2](#) [advance_window_bridges_fire_subphase_and_closes_at_melee](#) [fired_at_tracker_resets_at_maxim_subphase](#)

§6.51 – Leader Units

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:456 GH:omdurman-rules/src/lib.rs:456	BritishLeader	454 /// to claim the Mahdi's Tomb (§9.14). 455 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 456 pub enum BritishLeader { 457 Kitchener, 458 Gatacre,
omdurman-types/src/lib.rs:829 GH:omdurman-types/src/lib.rs:829	BritishLeader	827 DervishLeader { fire: i32, melee: i32, movement: i32 }, 828 /// Anglo-Egyptian leader (§6.51): movement only. 829 BritishLeader { movement: i32 }, 830 /// Wall-breach marker placed by artillery fire

		<pre> (\$6.63). Not a combat unit. 831 Breech, </pre>
omdurman-types/src/lib.rs:870 GH:omdurman-types/src/lib.rs:870	has_combat_factors	<pre> 868 /// British leaders print a movement factor only (\$6.51); other kinds carry 869 /// fire and/or melee factors. Markers carry no stats. 870 pub fn has_combat_factors(self) -> bool { 871 !matches!(self, UnitKind::BritishLeader { .. } UnitKind::Marker UnitKind::Breech UnitKind::BareCounter) 872 } </pre>

Covered by tests: [zero_factor_is_none_not_zero](#) [kitchener_block_resolves_leaders_friendlies_camel_and_sudanese](#) [dervish_leader_sections_resolve_leader_and_retinue_per_cell](#)

§6.52 – Anglo-Egyptian Friendlies Brigade

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:647 GH:omdurman-rules/src/lib.rs:647	is_friendlies	<pre> 645 /// "Friendlies" units obey several special rules (\$5.21, \$5.23, \$6.52, 646 /// \$9.14 victory conditions). 647 pub fn is_friendlies(&self) -> bool { 648 matches!(649 self, </pre>
omdurman-types/src/lib.rs:895 GH:omdurman-types/src/lib.rs:895	Friendlies	<pre> 893 /// Native volunteer brigade -- the Shaggyeh (\$6.52). Do not receive 894 /// brigade integrity (\$5.54 enumerates only British/Egyptian/Sudanese). 895 Friendlies, 896 } 897 </pre>

Covered by tests: [friendlies_counters_score_by_bank_not_as_leaders](#)

§6.53 – Royal Engineers demolition

[implemented](#)
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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:616 GH:omdurman-rules/src/lib.rs:616	RoyalEngineers	<pre> 614 /// The Royal Engineers (§6.53) -- a *specific* unit, not a class, so we 615 /// model it explicitly. 616 RoyalEngineers, 617 } 618 </pre>
omdurman-rules/src/lib.rs:807 GH:omdurman-rules/src/lib.rs:807	demolishing	<pre> 805 806 /// Volatile per-turn state of a unit -- disrupted, loaded onto a gunboat, 807 /// constructing the Zariba, demolishing a target, etc. (rulebook \$5, \$6). 808 /// 809 /// Multiple state flags can be in effect at once (e.g. a unit may be both </pre>
omdurman-rules/src/effects.rs:120 GH:omdurman-	Demolition	<pre> 118 119 /// Royal Engineers demolition (rulebook §6.53). 120 Demolition { </pre>

rules/src/ effects.rs:120		121 unit_id: UnitId, 122 target: DemolitionTarget,
omdurman-rules/src/effects.rs:3755 GH:omdurman-rules/src/effects.rs:3755	apply_demolition	3753 hex: HexCoord, 3754 participants: &[UnitId], 3755 paragraphs: Vec<String>, 3756) { 3757 let survivors: Vec<UnitId> = participants
omdurman-rules/src/lib.rs:1065 GH:omdurman-rules/src/lib.rs:1065	DemolitionTarget	1063 /// disrupted or driven off. 1064 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1065 pub enum DemolitionTarget { 1066 Fort(UnitId), 1067 WallHexside(HexsideRef),

Covered by tests: [demolition_cancelled_when_engineer_disrupted](#) [demolition_cancelled_when_engineer_moved_away](#)

§6.54 – Forts

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:871 GH:omdurman-rules/src/lib.rs:871	ZocReason	869 /// Used by the engine when answering "is this hex in an enemy ZOC?". 870 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 871 pub enum ZocReason { 872 /// Normal ZOC: any non-disrupted unit other than an Anglo-Egyptian 873 /// leader (§5.41) projects ZOC into each of its six adjacent hexes.
omdurman-rules/src/lib.rs:878 GH:omdurman-rules/src/lib.rs:878	Fort	876 GunboatVsGunboat, 877 /// Forts project ZOC out of, but not into, an empty fort (§5.44, §6.54). 878 Fort, 879 /// Walled-city ZOC: extends out through walls and gates but not in, 880 /// across a breach in both directions (§5.44).
omdurman-rules/src/lib.rs:835 GH:omdurman-rules/src/lib.rs:835	UnitState::may_act	833 impl UnitState { 834 /// A disrupted unit may not move, fire, or melee (rulebook §5, reference notes). 835 pub fn may_act(self) -> bool { 836 !self.disrupted 837 }
omdurman-rules/src/lib.rs:813 GH:omdurman-rules/src/lib.rs:813	UnitState	811 /// rather than one big enum. 812 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, Default)] 813 pub struct UnitState { 814 /// Reference table: "Disrupted units: no ZOC; may not move; may not fire 815 /// offensively or defensively; may not melee; are turned face up at the
omdurman-rules/src/lib.rs:969 GH:omdurman-rules/src/lib.rs:969	FireAttack	967 /// modifiers (rulebook §6). 968 #[derive(Serialize, Deserialize, Clone, Debug)] 969 pub struct FireAttack { 970 pub firing_player: Player, 971 pub phase: Phase,
omdurman-rules/src/lib.rs:984	FireAttack::net_modifier	982 impl FireAttack { 983 /// Sum of all fire modifiers applied to this attack (rulebook §6.24). 984 pub fn net_modifier(&self) -> i16 {

GH:omdurman-rules/src/lib.rs:984		<pre>985 self.modifiers.iter().map(m m.die_modifier()).sum() 986 }</pre>
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§6.61 – Only artillery may fire at gunboats; 3+ to sink

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Only artillery may fire at gunboats. A result of 3 or more on the combat results table is required to sink a gunboat. Any other result is a miss.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:513 GH:omdurman-rules/src/lib.rs:513	WeaponClass	<pre>511 /// enum so a "spear" unit cannot accidentally fire on the "Howitzer" line. 512 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 513 pub enum WeaponClass { 514 /// Dervish spears and swords -- no ranged fire at all. 515 Melee,</pre>

Covered by tests: [rifles_may_not_sink_a_gunboat](#)

§6.62 – Only artillery may fire at forts; 2+ to destroy

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Only artillery may fire at forts. A result of 2 or more on the combat results table is required to eliminate a fort. Any other result is a miss. If the fort contains any enemy units at the instant it is destroyed, one unit is eliminated with the fort.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:513 GH:omdurman-rules/src/lib.rs:513	WeaponClass	<pre>511 /// enum so a "spear" unit cannot accidentally fire on the "Howitzer" line. 512 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 513 pub enum WeaponClass { 514 /// Dervish spears and swords -- no ranged fire at all. 515 Melee,</pre>

§6.63 – Only artillery may breach wall hexsides; 2+ required

implemented
manual page unknown

Only artillery may fire to breach a wall hexside of Khartoum or the walled city of Omdurman. A result of 2 or more on the combat results table is required to breach a wall. Any other result is a miss. The effect of the breach is to negate the wall hexside for line of sight purposes. Place a "BREACH" marker in an adjacent hex so that the arrow points to the breached hexside. If any enemy units are adjacent to the wall hexside at the instant it is breached, one enemy unit is eliminated.

****6.64) Howitzer fire:**** Five units in the game have howitzer fire capability. These are the five named British gunboats. They may fire their artillery factor as direct fire during the Direct Fire Subphase (see 4 and 6.41) and may then fire the same artillery factor as howitzer fire during the Maxim Second Fire and Howitzer Subphase (see 4 and 6.42). Exception: no howitzer fire is allowed during night game turns. To fire howitzer fire, select any target hex between 4 and 10 hexes from the firing gunboat (ignoring the Line of Sight Table) and roll the ten-sided die twice. The first

die roll is the Combat Results Table die roll and the second roll is the impact hex die roll. Refer to the **Howitzer Fire** (see manual for full text)

See also: §6.41, §6.42, §6.64

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:187 GH:omdurman-types/src/lib.rs:187	Breach	<pre>185 /// Breach in a wall (artillery/§6.63 or Royal 186 Engineers/§6.53). ZOC both 187 /// ways; LOS no longer blocked across the hexside. 188 Breach, 189 /// Khor -- gully/wadi. ZOCs do not extend across 190 (§5.44); advance after 191 /// combat may not cross (§6.82).</pre>
omdurman-rules/src/effects.rs:216 GH:omdurman-rules/src/effects.rs:216	ArtilleryBreachWall	<pre>214 /// pre-rolled d10 used for the CRT lookup; range/ 215 LOS are re-derived by the 216 /// engine from the firers and `target`. 217 ArtilleryBreachWall { 218 firers: Vec<UnitId>, 219 target: HexsideRef,</pre>
omdurman-rules/src/effects.rs:3896 GH:omdurman-rules/src/effects.rs:3896	apply_artillery_breach_wall	<pre>3894 if *kind == HexsideKind::Wall { 3895 *kind = HexsideKind::Breach; 3896 } 3897 } else { 3898 state.board.hexsides.insert(edge, HexsideKind::Breach);</pre>
omdurman-rules/src/effects.rs:1735 GH:omdurman-rules/src/effects.rs:1735	can_fire_at_wall	<pre>1733 /// On success returns `(fire_factor, 1734       effective_range, nearer_endpoint)`. 1735 /// The caller is responsible for summing per- 1736 firer factors with the 1737 /// range band and resolving the CRT – this method 1738 only validates one 1739 /// firer at a time. 1740 pub fn can_fire_at_wall(</pre>

Covered by tests: [breach_marker_cell_returns_none](#)

§6.64 – Howitzer fire

[implemented](#)

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/howitzer_scatter.rs:6 GH:omdurman-rules/src/howitzer_scatter.rs:6	ScatterDirection	<pre>4 /// (§6.64). The caller maps these to hex-grid offsets. 5 #[derive(Clone, Copy, PartialEq, Eq, Debug)] 6 pub enum ScatterDirection { 7 /// Roll 7-10: hit the target hex. 8 OnTarget,</pre>
omdurman-rules/src/howitzer_scatter.rs:28 GH:omdurman-rules/src/howitzer_scatter.rs:28	howitzer_scatter	<pre>26 /// 3-4 [`ScatterDirection::Long`] (upstream) 27 /// 1-2 [`ScatterDirection::LeftRight`] 28 pub fn howitzer_scatter(impact_roll: DieRoll) -> ScatterDirection { 29 use DieRoll::*; 30 match impact_roll {</pre>
omdurman-rules/src/lib.rs:467 GH:omdurman-rules/src/lib.rs:467	GunboatId	<pre>465 /// fire; "old" gunboats do not (rulebook §2.32). 466 #[derive(Serialize, Deserialize, Clone, Copy, 467 PartialEq, Eq, Hash, Debug, strum::Display)] 468 pub enum GunboatId { 469 /// One of the five new-type named gunboats with 470 howitzer capability. 471 Named(NamedGunboat),</pre>
omdurman-rules/src/	HowitzerFire	<pre>67 68 /// Resolve a howitzer bombardment (two rolls:</pre>

effects.rs:69 GH:omdurman- rules/src/ effects.rs:69		Combat Results Table + impact scatter) (rulebook §6.64). 69 HowitzerFire { 70 attack: FireAttack, 71 combat_results_table_roll: DieRoll,
omdurman- rules/src/ effects.rs:2742 GH:omdurman- rules/src/ effects.rs:2742	apply_howitzer_fire	2740) 2741 } 2742 2743 /// Validate and apply a howitzer fire attack (scatter 2744 pub fn apply_howitzer_fire(path) (rulebook §6.64).
omdurman- rules/src/ lib.rs:1005 GH:omdurman- rules/src/ lib.rs:1005	HowitzerResolution	1003 /// roll on the Howitzer Fire Scattergram (§6.64). 1004 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1005 pub struct HowitzerResolution { 1006 pub combat_results_table_roll: DieRoll, 1007 pub impact_roll: DieRoll,
omdurman- rules/src/ lib.rs:1012 GH:omdurman- rules/src/ lib.rs:1012	HowitzerResolu- tion::hit_target_hex	1010 impl HowitzerResolution { 1011 /// The designated target hex is hit on impact roll 7-10 (§6.64). 1012 pub fn hit_target_hex(self) -> bool { 1013 use DieRoll::*; 1014 matches!(self.impact_roll, Seven Eight Nine Ten)
omdurman- rules/src/ effects.rs:1519 GH:omdurman- rules/src/ effects.rs:1519	can_fire_at	1517 /// hasn't already fired this phase, and the target is within (night- 1518 /// adjusted) range for the firer's weapon. 1519 /// 1520 /// Does **not** check line of sight or terrain -- those need the game map, 1521 /// which the rules engine does not hold; the app supplies the terrain

Covered by tests: howitzer_on_target_7_to_10
howitzer_scatters_below_7 howitzer_short_on_5_6 howitzer_long_on_3_4 howitzer_left_right_on_1_2 named_and_old_gunboats_resolve
named_gunboat_has_howitzer named_gunboat_may_fire_howitzer_in_second_subphase named_gunboat_direct_fire_uses_artillery_weapon
named_gunboat_no_howitzer_at_night dervish_gunboat_lacks_howitzer

§6.81 – Moving player may fire with all capable units

implicit
manual page unknown

During Offensive Fire phase, the moving player may fire with all of his units capable of firing, up to their maximum range, within the limitations imposed by the rules of combat.

§6.82 – Advance after combat (offensive fire)

implemented
manual page unknown

If an enemy-occupied hex is vacated as a result of offensive fire, friendly units may advance after combat into the vacated hex. To be eligible to advance, the friendly units must have participated in the attack and must have been adjacent to the vacated hex. Note that artillery may not advance, nor may units advance across a wall hexside (except at a gate or breach). Units may never advance after combat across a khor.

File	Symbol	Code Snippet
omdurman- rules/src/ effects.rs:107 GH:omdurman-	AdvanceAfterCombat	105 /// after fire, §7.6 after melee). Eligible units are adjacent attackers 106 /// that are not artillery; the target hex must be empty of enemies.

rules/src/ effects.rs:107		<pre>107 AdvanceAfterCombat { unit_id: UnitId, to: HexCoord }, 108 109 // -- Unit state changes -----</pre>
omdurman- rules/src/ effects.rs:2207 GH:omdurman- rules/src/ effects.rs:2207	apply_advance_after_comb	<pre>2205 GameEffect::RetreatBeforeMelee { unit_id, to } => { 2206 apply_retreat_before_melee(state, *unit_id, *to) 2207 } 2208 GameEffect::AdvanceAfterCombat { unit_id, to } => { 2209 apply_advance_after_combat(state, *unit_id, *to)</pre>
omdurman- types/src/ lib.rs:230 GH:omdurman- types/src/lib.rs:230	blocks_advance_after_comb	<pre>228 229 /// Whether advance-after-combat may *not* cross this side (§6.82, §7.6). 230 pub fn blocks_advance_after_combat(self) -> bool { 231 matches!(232 self,</pre>
omdurman- rules/src/ effects.rs:3374 GH:omdurman- rules/src/ effects.rs:3374	can_advance_after_combat	<pre>3372 } 3373 // Retreat is a *reaction* to a declared *infantry* melee attack on the 3374 // unit's hex (§7.5): there must be a pending melee targeting where it 3375 // stands, made by at least one infantry attacker. 3376 match &self.pending_melee {</pre>

Covered by tests: [can_advance_after_combat_rejects_wall_hexside](#) [can_advance_after_combat_rejects_khor_hexside](#)
[advance_requires_combat_vacated_hex](#) [advance_requires_participation](#)

7/7 implemented

§7 – Melee Phase

§7.1 – Melee strength printed on counter

implemented

manual page unknown

The melee strength of all units is printed on the counter. Note that gunboats have no melee strength. Gunboats may neither melee attack nor be melee attacked.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:118 GH:omdurman-rules/src/lib.rs:118	MeleeFactor	<pre>116 /// Every possible value from the annotated counter set is a named variant. 117 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 118 pub enum MeleeFactor { 119 One = 1, 120 Three = 3,</pre>
omdurman-rules/src/lib.rs:129 GH:omdurman-rules/src/lib.rs:129	MeleeFactor::sum	<pre>127 impl MeleeFactor { 128 /// Sum multiple melee factors (rulebook §7.1). 129 pub fn sum<'a>(factors: impl IntoIterator<Item = &'a MeleeFactor>) -> u16 { 130 factors.into_iter().map(f f.value()).sum() 131 }</pre>
omdurman-types/src/lib.rs:853 GH:omdurman-types/src/lib.rs:853	may_be_melee_attacked	<pre>851 852 /// Gunboats neither attack nor are attacked in melee (§7.1). 853 pub fn may_be_melee_attacked(self) -> bool { 854 !matches!(self, UnitKind::Gunboat { .. }) 855 }</pre>

§7.2 – Melee adjacent only, not across wall hexsides

implemented

manual page unknown

Melee simulates the hand-to-hand fighting of the period. Units may melee attack adjacent enemy units only. Units may not melee attack across a wall hexside, but may melee attack through a gate or breach hexside.

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:225 GH:omdurman-types/src/lib.rs:225	blocks_melee	<pre>223 /// Whether melee may *not* be made across this side (§7.2). Gates and 224 /// breaches are passable to melee. 225 pub fn blocks_melee(self) -> bool { 226 matches!(self, HexsideKind::Wall HexsideKind::ZaribaThornHedge) 227 }</pre>
omdurman-rules/src/effects.rs:1846 GH:omdurman-rules/src/effects.rs:1846	can_melee	<pre>1844 /// active player, attacker is a melee-capable kind (§7.4), not disrupted, 1845 /// adjacent to the target, the target hex holds at least one enemy unit 1846 /// that may be melee-attacked (gunboats may not -- §7.1), and no wall or 1847 /// thorn-hedge hexside blocks the attack (§7.2). 1848 pub fn can_melee(&self, attacker: UnitId, defender_hex: HexCoord) -> Result<(), RuleError> {</pre>

Covered by tests: [can_melee_gates_phase_adjacency_and_kind](#) [can_melee_rejects_wall_hexside](#) [can_melee_rejects_thorn_hedge_hexside](#) [can_melee_allows_gate_hexside](#)

§7.3 – Simultaneous melee combat

implemented

manual page unknown

Melee combat is considered simultaneous, so that units eliminated by melee attacks still get a melee combat die roll.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:79 GH:omdurman-rules/src/effects.rs:79	MeleeCombat	<pre>77 /// Used for an immediate resolution with no 78 /// resolution primitive in tests). 79 MeleeCombat { 80 attack: MeleeAttack, 81 attacker_roll: DieRoll,</pre>
omdurman-rules/src/effects.rs:3064 GH:omdurman-rules/src/effects.rs:3064	apply_melee_combat	<pre>3062 { 3063 open_advance_window(3064 state, 3065 target_hex, 3066 &attack.firers,</pre>

§7.4 – Who may melee attack / defend

implemented

manual page unknown

Only infantry, cavalry, camel units, and Dervish leaders may melee attack. All units (except gunboats — see 7.1) may melee defend.

See also: §7.1

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:842 GH:omdurman-types/src/lib.rs:842	may_melee_attack	<pre>840 /// Rulebook §7.4 -- only infantry, cavalry, camel 841 /// and Dervish leaders may 842 /// melee *attack*. All others (except gunboats) 843 /// may melee *defend* (§7.1). 844 pub fn may_melee_attack(self) -> bool { 845 matches!(846 self,</pre>
omdurman-types/src/lib.rs:807 GH:omdurman-types/src/lib.rs:807	UnitKind	<pre>805 /// `Some(UnitKind::Marker)` or `None`. 806 #[derive(Serialize, Deserialize, Clone, Copy, 807 PartialEq, Eq, Hash, Debug, strum::Display)] 808 pub enum UnitKind { 809 /// Foot infantry (§2.3): fire / melee / movement. 810 Infantry { fire: i32, melee: i32, movement: i32 },</pre>
omdurman-types/src/lib.rs:664 GH:omdurman-types/src/lib.rs:664	DervishTribe	<pre>662 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, 663 Hash, Debug, strum::Display, strum::EnumIter, 664)] 665 pub enum DervishTribe { 666 Baggara, 667 Jaalin,</pre>
omdurman-rules/src/effects.rs:1846 GH:omdurman-rules/src/effects.rs:1846	can_melee	<pre>1844 /// active player, attacker is a melee-capable 1845 /// kind (§7.4), not disrupted, 1846 /// adjacent to the target, the target hex holds 1847 /// at least one enemy unit 1848 /// that may be melee-attacked (gunboats may not 1849 /// -- §7.1), and no wall or 1850 /// thorn-hedge hexside blocks the attack (§7.2). 1851 pub fn can_melee(&self, attacker: UnitId, 1852 defender_hex: HexCoord) -> Result<(), RuleError> {</pre>

§7.5 – Cavalry/camel retreat before melee

implemented

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Cavalry and camel units may retreat two hexes from an infantry melee attack. Note, however, that only one retreat per unit per turn is permitted. Thus, if their retreat places them adjacent to enemy units whose melee attacks have not yet been resolved, those enemy units may elect to attack the retreating unit(s).

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:102 GH:omdurman-rules/src/effects.rs:102	RetreatBeforeMelee	<pre>100 /// melee attack, *before* it is resolved (§7.5). One retreat per unit per 101 /// turn. (rulebook §7.5). 102 RetreatBeforeMelee { unit_id: UnitId, to: HexCoord }, 103 104 /// An attacking unit advances into a hex vacated by combat (rulebook §6.82</pre>
omdurman-rules/src/effects.rs:2204 GH:omdurman-rules/src/effects.rs:2204	apply_retreat_before_melee	<pre>2202 defender_roll, 2203 } => apply_declare_melee(state, attack, *attacker_roll, *defender_roll), 2204 GameEffect::ResolveMelee => apply_resolve_melee(state), 2205 GameEffect::RetreatBeforeMelee { unit_id, to } => { 2206 apply_retreat_before_melee(state, *unit_id, *to)</pre>
omdurman-types/src/lib.rs:859 GH:omdurman-types/src/lib.rs:859	may_retreat_before_melee	<pre>857 /// Cavalry and camel units may retreat two hexes from an infantry melee 858 /// attack (§7.5). 859 pub fn may_retreat_before_melee(self) -> bool { 860 matches!(self, UnitKind::Cavalry { .. } UnitKind::Camel { .. }) 861 }</pre>
omdurman-rules/src/lib.rs:195 GH:omdurman-rules/src/lib.rs:195	HexDistance	<pre>193 /// (rulebook §6.22, §7.5). 194 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 195 pub struct HexDistance(u16); 196 197 impl HexDistance {</pre>
omdurman-rules/src/effects.rs:3328 GH:omdurman-rules/src/effects.rs:3328	can_retreat_before_melee	<pre>3326 .filter(u { 3327 u.position == attack.defender_hex 3328 && u.profile.identity.owner() == defender_player 3329 && u.profile.kind.may_be_melee_attacked() 3330 })</pre>

Covered by tests: [retreat_before_melee_only_cavalry_two_hexes](#) [retreat_opens_window_only_when_hex_emptyies](#)

§7.6 – Advance after melee

implemented

manual page unknown

If a melee attack eliminates all of the defenders in an adjacent hex, the Dervish player MUST advance into the vacated hex. To be eligible to advance, the Dervish units must have been adjacent to the vacated hex and participated in the melee attack that eliminated the defenders. All surviving eligible Dervish units MUST advance, up to the stacking limit. The Anglo-Egyptian player may advance if desired. Note that only attacking units may advance.

File	Symbol	Code Snippet
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omdurman-rules/src/effects.rs:107 GH:omdurman-rules/src/effects.rs:107	AdvanceAfterCombat	<pre> 105 /// after fire, §7.6 after melee). Eligible units are adjacent attackers 106 /// that are not artillery; the target hex must be empty of enemies. 107 AdvanceAfterCombat { unit_id: UnitId, to: HexCoord }, 108 109 // -- Unit state changes ----- </pre>
omdurman-rules/src/effects.rs:2207 GH:omdurman-rules/src/effects.rs:2207	apply_advance_after_comb	<pre> 2205 GameEffect::RetreatBeforeMelee { unit_id, to } => { 2206 apply_retreat_before_melee(state, *unit_id, *to) 2207 } 2208 GameEffect::AdvanceAfterCombat { unit_id, to } => { 2209 apply_advance_after_combat(state, *unit_id, *to) </pre>
omdurman-rules/src/effects.rs:3374 GH:omdurman-rules/src/effects.rs:3374	can_advance_after_combat	<pre> 3372 } 3373 // Retreat is a *reaction* to a declared *infantry* melee attack on the 3374 // unit's hex (§7.5): there must be a pending melee targeting where it 3375 // stands, made by at least one infantry attacker. 3376 match &self.pending_melee { </pre>

§7.7 – Melee modifiers

implemented

manual page unknown

To resolve melee, both the attacker and the defender roll on the Combat Results Table and apply the applicable melee modifier to their die roll. The Dervish player receives a +2 melee modifier, the Anglo-Egyptian player receives a +1 melee modifier. No terrain modifiers are applied to melee combat (Exception: Zariba hexsides in the historical scenario and the campaign game, if constructed — see 9.23). Melee losses must be taken from meleeing units first!

See also: §9.23

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1023 GH:omdurman-rules/src/lib.rs:1023	MeleeModifier	<pre> 1021 1022 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1023 pub enum MeleeModifier { 1024 /// +2 to all Dervish melee rolls (§7.7). 1025 DervishStandard, </pre>
omdurman-rules/src/lib.rs:241 GH:omdurman-rules/src/lib.rs:241	DieModifier	<pre> 239 /// A die-roll modifier from a single named source (rulebook §6.24, §7.7). 240 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, Default)] 241 pub enum DieModifier { 242 #[default] 243 Zero, </pre>
omdurman-rules/src/lib.rs:1046 GH:omdurman-rules/src/lib.rs:1046	MeleeAttack	<pre> 1044 /// A melee attack: simultaneous, both sides roll on the Combat Results Table (§7.3, §7.7). 1045 #[derive(Serialize, Deserialize, Clone, Debug)] 1046 pub struct MeleeAttack { 1047 pub attacker_player: Player, 1048 pub attacker_hex: HexCoord, </pre>
omdurman-rules/src/lib.rs:1027 GH:omdurman-	MeleeModifier::AngloE- gyptianStandard	<pre> 1025 DervishStandard, 1026 /// +1 to all Anglo-Egyptian melee rolls (§7.7). 1027 AngloEgyptianStandard, 1028 /// Inverted to -2 when Dervish units melee-attack </pre>

rules/src/ lib.rs:1027		<pre>across a trench into 1029 /// an entrenched defender (§9.232).</pre>
omdurman-rules/src/lib.rs:1030 GH:omdurman-rules/src/lib.rs:1030	MeleeModifier::DervishVsTrenchDefender	<pre>1028 /// Inverted to -2 when Dervish units melee-attack across a trench into 1029 /// an entrenched defender (§9.232). 1030 DervishVsTrenchedDefender, 1031 } 1032 </pre>
omdurman-rules/src/lib.rs:1025 GH:omdurman-rules/src/lib.rs:1025	MeleeModifier::DervishStandard	<pre>1023 pub enum MeleeModifier { 1024 /// +2 to all Dervish melee rolls (§7.7). 1025 DervishStandard, 1026 /// +1 to all Anglo-Egyptian melee rolls (§7.7). 1027 AngloEgyptianStandard,</pre>

Covered by tests: [melee_resolves_simultaneously](#)

■ 2/2 implemented

§8 – Night Game Turns

§8.1 – Night effects

implemented

manual page unknown

The effects of night game turns are: a) all Anglo-Egyptian movement allowances are halved (round down), b) there is no Anglo-Egyptian howitzer fire, and c) all fire ranges are halved for both sides (round down, but range 1 stays range 1). Range effects on fire combat are the same as during day game turns. For example, an Anglo-Egyptian infantry unit firing at night will be doubled at range 1, normal at range 2, and may not fire at range 3 or greater.

****8.2) Dervish Desertion Roll:**** Once each campaign game, during the first night turn of the game, the Dervish player rolls one die to see how many of his units desert. The roll is made during the movement phase and the number of deserting Dervish units is equal to 1½ times the roll of one die. The Dervish player may choose which units desert by merely removing them from the mapsheet. The KHALIFA unit, gunboats, artillery units, and forts are the only Dervish units that may not be chosen. No victory points are awarded to the Anglo-Egyptian player for deserting Dervishes.

(see manual for full text)

See also: §8.2

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:164 GH:omdurman-rules/src/lib.rs:164	MovementAllowance::halve	<pre>162 impl MovementAllowance { 163 /// Night movement allowance = halved (round down) (rulebook §8.1, §5.11). 164 pub fn halve(self) -> Self { 165 let v = self.value() / 2; 166 MovementAllowance::try_from(v).expect("halved value always a named variant")</pre>
omdurman-types/src/lib.rs:727 GH:omdurman-types/src/lib.rs:727	DayNight	<pre>725 /// (rulebook §8.1). 726 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 727 pub enum DayNight { 728 Day, 729 Night,</pre>
omdurman-rules/src/range_effects.rs:116 GH:omdurman-rules/src/range_effects.rs:116	night_max_range	<pre>114 115 /// The halved maximum range at night (§8.1): round down, minimum 1. 116 pub fn night_max_range(weapon: WeaponClass, ae: bool) - > u8 { 117 let day = max_day_range(weapon, ae); 118 if day <= 1 { 1 } else { day / 2 }</pre>
omdurman-rules/src/lib.rs:1518 GH:omdurman-rules/src/lib.rs:1518	effective_movement_at_night	<pre>1516 /// Apply night-turn movement halving for Anglo- Egyptian units (§8.1): all 1517 /// Anglo-Egyptian movement allowances are halved (round down). 1518 pub fn effective_movement_at_night(1519 allowance: MovementAllowance, 1520 player: Player,</pre>

Covered by tests: [night_max_ranges](#) [night_max_ranges_remaining](#) [ae_rifle_at_night_matches_rulebook_example](#) [max_day_range_all_combos](#) [night_movement_overlay_allowance_halved](#)

§8.2 – Dervish Desertion Roll

implemented

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File	Symbol	Code Snippet
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omdurman-rules/src/effects.rs:135 GH:omdurman-rules/src/effects.rs:135	DervishDesertion	<pre> 133 /// the effect. The Khalifa, gunboats, artillery, and forts may not be 134 /// chosen. 135 DervishDesertion { 136 roll: DieRoll, 137 deserters: Vec<UnitId>, </pre>
omdurman-rules/src/turn_track.rs:63 GH:omdurman-rules/src/turn_track.rs:63	DervishDesertion	<pre> 61 None, 62 /// Dervish desertion roll (§8.2) -- occurs on the first night turn. 63 DervishDesertion, 64 /// Dervish reinforcements are available. 65 DervishReinforcements, </pre>
omdurman-rules/src/turn_track.rs:60 GH:omdurman-rules/src/turn_track.rs:60	TurnEvent	<pre> 58 /// Special events that occur on specific turns (rulebook §8.2, §9.112, §9.113). 59 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 60 pub enum TurnEvent { 61 None, 62 /// Dervish desertion roll (§8.2) -- occurs on the first night turn. </pre>

Covered by tests: [desertion_count_is_floor_one_and_a_half](#) [desertion_on_first_night](#)

 22/32 implemented

§9 – The Scenarios

§9.1 – The Campaign Game

descriptive

manual page unknown

The Campaign Game

Covered by tests: `campaign_has_no_fixed_placements` `scenario_maps_to_board`

§9.2 – The Historical Scenario

descriptive

manual page unknown

The Historical Scenario

Players should note that the historical scenario is an exercise in futility for the Dervish player. It is, however, an interesting demonstration of the absolute imbecility of the Khalifa’s generalship and vividly shows the superiority of entrenched firepower over traditional tribal arms in the colonial period.

Covered by tests: `start_game_scenario_selects_board` `remove_deployed_unit_happy_path`

§9.3 – Bonus Game: Fall of Khartoum

out-of-scope

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Bonus Game: FALL OF KHARTOUM

Covered by tests: `remove_deployed_unit_happy_path`

§9.11 – Set Up (Campaign)

out-of-scope

manual page unknown

Set Up

§9.12 – Scenario Length (Campaign)

implemented

manual page unknown

Scenario Length

6:00 am, Sept. 1 through 8:00 am, Sept. 3, 22 Game Turns.

File	Symbol	Code Snippet
<code>omdurman-rules/src/turn_track.rs:11</code> GH:omdurman-rules/src/turn_track.rs:11	<code>GameTime</code>	<pre>9 /// starts at one of these twelve times. 10 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 11 pub enum GameTime { 12 SixAM, 13 EightAM,</pre>

omdurman-rules/src/turn_track.rs:47 GH:omdurman-rules/src/turn_track.rs:47	TurnEntry	<pre>45 /// A single entry on the Turn Record Track (rulebook §9.12, §9.22). 46 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 47 pub struct TurnEntry { 48 /// 1-based turn number. 49 pub turn: u8,</pre>
omdurman-rules/src/turn_track.rs:95 GH:omdurman-rules/src/turn_track.rs:95	CAMPAIGN_TURN_TRACK	<pre>93 /// is turn 9, which carries the once-per-game Dervish Desertion Roll (§8.2) -- 94 /// the printed track prints "Dervish Desertion Roll / NIGHT" on that cell. 95 pub const CAMPAIGN_TURN_TRACK: [TurnEntry; 22] = [96 // Row 1, left->right: Sept 1, 6 am -> 8 pm, then the first NIGHT. 97 entry(1, SixAM, Day, TurnEvent::None),</pre>
omdurman-rules/src/turn_track.rs:242 GH:omdurman-rules/src/turn_track.rs:242	TurnLabel	<pre>240 /// printed cell; `Blank` is for unused positions in the 9x3 grid. 241 #[derive(Serialize, Deserialize, Debug, Clone, Copy, PartialEq, Eq)] 242 pub enum TurnLabel { 243 Blank, 244 /// "SEPT. 1\n6:00 am" -- day header plus time (turn 1).</pre>
omdurman-rules/src/turn_track.rs:340 GH:omdurman-rules/src/turn_track.rs:340	turn_marker_pixel	<pre>338 /// 339 /// Rows 0 and 1 use all 9 columns; row 2 uses only columns 0-3. 340 pub fn turn_marker_pixel(track: &omdurman_types::CampaignTurnTrack, turn: u8) -> (f32, f32) { 341 let cell_w = track.w / 9.0; 342 let cell_h = track.h / 3.0;</pre>

Covered by tests: [campaign_track_22_turns](#) [desertion_on_first_night](#) [campaign_track_label_and_day_night_agree](#) [game_time_display_all_variants](#) [turn_label_display](#) [turn_label_out_of_range_is_none](#) [turn_marker_pixel_row_0_left_to_right](#) [turn_marker_pixel_row_1_right_to_left](#) [turn_marker_pixel_rows_are_stacked](#) [game_over_after_campaign_turns](#)

§9.13 – Special Rules (Campaign)

[descriptive](#)
manual page unknown

Special Rules

None.

§9.14 – Victory Conditions (Campaign)

[implemented](#)
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Victory Conditions

The Mahdi’s Tomb in Omdurman was not only the tallest structure in the entire Sudan in 1898, it was also a Dervish holy shrine. Its loss or destruction would be a severe blow to the Mahdist cause. It is accordingly assigned 25 victory points which are awarded to the player who controls it at the conclusion of play. The Dervish player controls it at the start of play. As a tactical note, the Anglo-Egyptian player will find a decisive victory almost impossible unless he takes the Mahdi’s Tomb from the Dervish player. To take the Tomb hex, it must be occupied by one British leader plus any one non-“Friendlies” Anglo-Egyptian combat unit (both undisrupted) at the conclusion of play.

Additional victory points are awarded as follows:

****Dervish Player receives:****

- 10 pts: each British leader eliminated

(see manual for full text)

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1164 GH:omdurman-rules/src/lib.rs:1164	VpSource	<pre>1162 /// the manual and the engine. 1163 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1164 pub enum VpSource { 1165 // ----- Anglo-Egyptian player receives: 1166 /// Mahdi's Tomb control at conclusion of play (\$9.14).</pre>
omdurman-rules/src/lib.rs:1190 GH:omdurman-rules/src/lib.rs:1190	VpSource::points	<pre>1188 impl VpSource { 1189 /// VP awarded to `who_scores()` (rulebook §9.14). 1190 pub fn points(self) -> VictoryPoints { 1191 match self { 1192 VpSource::MahdisTomb => VictoryPoints::new(25),</pre>
omdurman-rules/src/lib.rs:1205 GH:omdurman-rules/src/lib.rs:1205	VpSource::who_scores	<pre>1203 1204 /// Which player receives these victory points (rulebook §9.14). 1205 pub fn who_scores(self) -> Player { 1206 match self { 1207 VpSource::MahdisTomb</pre>
omdurman-rules/src/lib.rs:1240 GH:omdurman-rules/src/lib.rs:1240	VictoryLedger	<pre>1238 /// Cumulative victory ledger for one scenario (rulebook §9.14). 1239 #[derive(Serialize, Deserialize, Clone, Debug, Default)] 1240 pub struct VictoryLedger { 1241 pub events: Vec<VpEvent>, 1242 }</pre>
omdurman-rules/src/lib.rs:1246 GH:omdurman-rules/src/lib.rs:1246	VpEvent	<pre>1244 /// A single victory-point scoring event (rulebook §9.14). 1245 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1246 pub struct VpEvent { 1247 pub turn: GameTurnIndex, 1248 pub source: VpSource,</pre>
omdurman-rules/src/lib.rs:1253 GH:omdurman-rules/src/lib.rs:1253	VictoryLedger::total_for	<pre>1251 impl VictoryLedger { 1252 /// Total victory points earned by a given player (rulebook §9.14). 1253 pub fn total_for(&self, player: Player) -> VictoryPoints { 1254 VictoryPoints(1255 self.events</pre>
omdurman-rules/src/lib.rs:1265 GH:omdurman-rules/src/lib.rs:1265	VictoryLedger::superiority	<pre>1263 /// Net superiority: positive = Anglo-Egyptian ahead, negative = Dervish ahead 1264 /// (rulebook §9.14). 1265 pub fn superiority(&self) -> VictoryPoints { 1266 VictoryPoints(self.total_for(Player::AngloEgyptian).value() - self.total_for(Player::Dervish).value()) 1267 }</pre>
omdurman-rules/src/lib.rs:1283 GH:omdurman-rules/src/lib.rs:1283	CampaignVictoryLevel	<pre>1281 /// Campaign-game victory levels (§9.14). 1282 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1283 pub enum CampaignVictoryLevel { 1284 Draw, 1285 Marginal(Player),</pre>
omdurman-rules/src/lib.rs:1292 GH:omdurman-rules/src/lib.rs:1292	CampaignVictoryLevel::from	<pre>1290 impl CampaignVictoryLevel { 1291 /// Assign a level from the net superiority (§9.14). 1292 pub fn from_superiority(s: VictoryPoints) -> Self { 1293 let net = s.0;</pre>

		1294 // Positive -> Anglo-Egyptian thresholds: 15/30/50
omdurman-rules/src/effects.rs:3842 GH:omdurman-rules/src/effects.rs:3842	score_elimination	3840 /// adjacent at the instant of breaching, one is eliminated. 3841 /// 3842 /// Either way the engineer is freed (`demolishing = false`). 3843 pub fn apply_resolve_demolition(3844 state: &mut GameState,
omdurman-rules/src/lib.rs:270 GH:omdurman-rules/src/lib.rs:270	VictoryPoints	268 /// (rulebook §9.14). 269 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Debug, Default)] 270 pub struct VictoryPoints(i32); 271 272 impl VictoryPoints {

Covered by tests:

friendlies_bank_scores_by_side

mahdis_tomb_not_scored_without_a_leader

mahdis_tomb_scores_for_anglo_egyptian_when_held

vp_source_attributes

§9.21 – Set Up (Historical)

out-of-scope

manual page unknown

Set Up

§9.22 – Scenario Length (Historical)

implemented

manual page unknown

Scenario Length

6:00 am, September 2 through 12:00 noon, September 2. Four game turns.

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:132 GH:omdurman-rules/src/turn_track.rs:132	HISTORICAL_TURN_T	130 131 /// Historical scenario track (§9.22 -- 4 turns, Sept 2 6:00 am -> 12:00 pm). 132 pub const HISTORICAL_TURN_TRACK: [TurnEntry; 4] = [133 TurnEntry { 134 turn: 1,

Covered by tests:

historical_turn_all_four_turns

§9.23 – Special Rule: The Zariba

implemented

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Special Rule: “The Zariba”

****9.231)** Thorn hedge hexsides:****** –2 to die roll on all Dervish fire attacks; may not melee across in either direction; may not advance after combat across in either direction.

****9.232)** Trench hexsides:****** –4 to die roll on all Dervish fire attacks vs. entrenched units only; –2 (instead of +2) melee modifier to Dervish units melee attacking an entrenched unit; entrenched units may be fired “over” in both directions (i.e. they do not block line of sight); units are considered “entrenched” if they are directly adjacent to (and on the Nile River side of) a trench hexside.

(see manual for full text)

See also: §9.231, §9.232

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:932 GH:omdurman-rules/src/lib.rs:932	ZaribaThornHedge	<pre>930 Terrain(i16), 931 /// -2 thorn-hedge defensive modifier (§9.231). 932 ZaribaThornHedge, 933 /// -4 trench defensive modifier (§9.232). Only applies vs. "entrenched" 934 /// units (those Nile-side of the trench hexside).</pre>
omdurman-rules/src/lib.rs:935 GH:omdurman-rules/src/lib.rs:935	ZaribaTrenchEntrenched	<pre>933 /// -4 trench defensive modifier (§9.232). Only applies vs. "entrenched" 934 /// units (those Nile-side of the trench hexside). 935 ZaribaTrenchEntrenched, 936 } 937 </pre>
omdurman-types/src/lib.rs:195 GH:omdurman-types/src/lib.rs:195	ZaribaThornHedge	<pre>193 Crest, 194 /// Historical-scenario thorn-hedge segment of the Zariba (§9.231). 195 ZaribaThornHedge, 196 /// Historical-scenario trench segment of the Zariba (§9.232). 197 ZaribaTrench,</pre>
omdurman-types/src/lib.rs:197 GH:omdurman-types/src/lib.rs:197	ZaribaTrench	<pre>195 ZaribaThornHedge, 196 /// Historical-scenario trench segment of the Zariba (§9.232). 197 ZaribaTrench, 198 /// One of the two end hexsides of a Zariba trench segment that connect to 199 /// the Nile River (§9.233). Units may only enter/ leave the Zariba via</pre>

§9.24 – Victory Conditions (Historical)

implemented

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Victory Conditions

Victory Levels are based solely on eliminating enemy units while conserving your own force as much as possible.

| Victory Level | Anglo-Egyptian Player (Dervish units eliminated) | Dervish Player (Anglo-Egyptian units eliminated)
| |—|—|—| | 5 — DECISIVE | 100+ | 30+ | | 4 — STRATEGIC | 60–99 | 15–29 | | 3 — TACTICAL | 45–59 | 10–14 | | 2 — MARGINAL | 30–44 | 5–9 | | 1 — DRAW | 0–29 | 0–4 |

The lower value victory level is then subtracted from the higher level to determine a player's net victory. For example, if the Anglo-Egyptian player eliminates 104 Dervish units (decisive victory) but loses 18 units doing it (Dervish Strategic), the Anglo-Egyptian player only nets out with a draw (decisive worth 5 minus strategic worth 4 = 1, draw).

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1322 GH:omdurman-rules/src/lib.rs:1322	HistoricalVictoryLevel	<pre>1320 /// draw"). 1321 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Debug)] 1322 pub enum HistoricalVictoryLevel { 1323 Draw = 1, 1324 Marginal = 2,</pre>

Covered by tests: `historical_victory_level_for_dervish` `historical_victory_level_for_anglo_egyptian`

§9.31 – Bonus game map

`out-of-scope`
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Only the small FALL OF KHARTOUM scenario map is used for this game.

§9.32 – Set Up (Bonus)

`out-of-scope`
manual page unknown

Set Up

§9.33 – Scenario Length (Bonus)

`implemented`
manual page unknown

Scenario Length
Variable, see victory conditions (9.35). Rarely lasts five turns.

See also: §9.35

File	Symbol	Code Snippet
<code>omdurman-rules/src/turn_track.rs:172</code> GH:omdurman-rules/src/turn_track.rs:172	<code>FALL_OF_KHARTOUM</code>	<pre>170 /// (the rulebook fixes none); only `day_night` is rule-bearing (night halves 171 /// Anglo-Egyptian movement and ranges and bars howitzer fire, §8.1). 172 pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] = [173 TurnEntry { 174 turn: 1,</pre>

Covered by tests: `fall_of_khartoum_turn_one_is_night` `fall_of_khartoum_turns_3_to_8_are_day`

§9.34 – Special Rules (Bonus)

`out-of-scope`
manual page unknown

Special Rules

§9.35 – Victory Conditions (Bonus)

`implemented`
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Victory Conditions

Victory is determined by how many turns it takes the Dervish player to eliminate the GORDON leader unit and how many Dervish units are eliminated:

- Dervish decisive: eliminate GORDON turn four or sooner.
- Dervish tactical: eliminate GORDON turn five.
- Dervish marginal: eliminate GORDON turn six.

- British marginal: GORDON survives end of turn six.
- British tactical: GORDON survives end of turn seven.
- British decisive: GORDON survives end of turn eight.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1363 GH:omdurman-rules/src/lib.rs:1363	FoKVictoryLevel	<pre> 1361 /// negative) so the loss penalty is a simple shift toward the British end. 1362 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Debug)] 1363 pub enum FoKVictoryLevel { 1364 DervishDecisive = -3, 1365 DervishTactical = -2,</pre>

Covered by tests: [fok_victory_level_worked_example](#) [fok_victory_level_gordon_died_early](#) [fok_victory_level_late_gordon_death](#) [fok_victory_level_gordon_survived](#)

§9.111 – Dervish set up (Campaign)

[implemented](#)

manual page unknown

Dervish player sets up first, moves second.

- Isa Zachneih infantry unit: anywhere on the east bank, in or south of El Debeba.
- KHALIFA ABDULLAH: in the walled city of Omdurman, in either palace hex.
- 3 artillery units, and all Taiasha units: anywhere in the walled city of Omdurman.
- 17 forts: anywhere on the mapsheet south of the Khor Shambat on the west bank, and/or south of all Halfaya hut hexes on the east bank and Nile River islands.
- 2 gunboats: any south edge Nile River hexes.

****9.112) Dervish reinforcements:**** all reinforcements enter on the west edge of the mapsheet, south of the Khor Shambat. Each unit pays the terrain cost of the hex through which it enters, no matter how many units enter through

(see manual for full text)

See also: §9.112

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:900 GH:omdurman-rules/src/effects.rs:900	setup_complete	<pre> 898 /// engine's `BoardInfo` carries no 00B); river mines/chain within limits are 899 /// enforced at placement time, so they need no re- check here. 900 /// 901 /// Returns [`RuleError::SetupIncomplete`] naming the first unmet requirement, 902 /// so the UI can surface *why* "Begin battle" is disabled. Every scenario</pre>

Covered by tests: [hadendowa_first_cell_is_isa_zachneih](#) [campaign_setup_rejects_non_initial_force](#)

§9.112 – Dervish reinforcements (Campaign)

[implemented](#)

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File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:127 GH:omdurman-	PlaceReinforcements	<pre> 125 // -- Reinforcement / placement ----- 126 /// Place reinforcements onto the map (rulebook §9.112, §9.113).</pre>

rules/src/ effects.rs:127		<pre> 127 PlaceReinforcements(Vec<UnitPlacement>), 128 129 // -- Scenario-specific ----- </pre>
omdurman- rules/src/ effects.rs:2215 GH:omdurman- rules/src/ effects.rs:2215	apply_place_reinforcement	<pre> 2213 apply_construct_zariba(state, unit_ids, *hexside) 2214 } 2215 GameEffect::Demolition { unit_id, target } => apply_demolition(state, *unit_id, *target), 2216 GameEffect::PlaceReinforcements(placements) => { 2217 apply_place_reinforcements(state, placements) </pre>
omdurman- rules/src/reinforcements.rs:74 GH:omdurman- rules/src/reinforcements.rs:74	dervish_campaign_schedule	<pre> 72 /// All reinforcements enter on the west edge, south of the Khor Shambat. 73 /// Each unit pays terrain cost of the hex it enters through. 74 pub fn dervish_campaign_schedule() -> ReinforcementSchedule { 75 ReinforcementSchedule { 76 player: Player::Dervish, </pre>
omdurman- types/src/ lib.rs:509 GH:omdurman- types/src/lib.rs:509	Location	<pre> 507 /// Named map landmarks (rulebook mapsheet, §9.111, §9.113, §9.212 scenarios). 508 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)] 509 pub enum Location { 510 FortMakran, 511 NorthFort, </pre>
omdurman- types/src/ lib.rs:589 GH:omdurman- types/src/lib.rs:589	SetupLetter	<pre> 587 /// Each letter marks a specific hex where a Dervish leader is placed. 588 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)] 589 pub enum SetupLetter { 590 Y, 591 K, </pre>
omdurman- types/src/ lib.rs:687 GH:omdurman- types/src/lib.rs:687	Faction	<pre> 685 /// `Some(BrigadeId::friendlies())`. 686 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 687 pub enum Faction { 688 Dervish { tribe: DervishTribe }, 689 BritishEgyptian { </pre>

Covered by tests: `dervish_schedule_has_three_waves` `dervish_wave_one_has_baggaara_and_three_leaders` `dervish_wave_two_has_hadendowa` `dervish_wave_three_is_all_remaining` `wave_for_turn_returns_correct_wave` `campaign_reinforcements_gate_by_wave`

§9.113 – Anglo-Egyptian reinforcements (Campaign)

implemented

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The Anglo-Egyptian player moves first. There are no Anglo-Egyptian units on the mapsheet at start. The GORDON unit is not used in this scenario.

- The leader units KITCHENER, GATACRE, and HUNTER may enter anytime during the first four game turns and do not count against the 12 units per turn limit. All three leaders must be in play by the end of turn four!
- All gunboats enter through any north edge Nile River hex, paying one movement point for the first hex entered. The “Friendlies” brigade enters through the Abu Alim hut hex on the east bank, paying eight movement points per unit. All other Anglo-Egyptian units enter through the west bank “ANGLO-EGYPTIAN ENTRANCE AREA”, each unit paying one movement point to enter the mapsheet.
- Turn 1) Any three gunboats; “Friendlies” brigade; Egyptian Cavalry; Horse Artillery; and two infantry brigades

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/reinforcements.rs:131 GH:omdurman-rules/src/reinforcements.rs:131	anglo_egyptian_campaign	<pre> 129 /// - "Friendlies" enter via Abu Alim hut on the east bank (8 MP per unit). 130 /// - All other AE units enter via the Anglo-Egyptian Entrance Area (1 MP). 131 pub fn anglo_egyptian_campaign_schedule() -> ReinforcementSchedule { 132 let free_leaders = vec![133 CampaignLeader::British(BritishLeader::Kitchener), </pre>

Covered by tests: `anglo_egyptian_schedule_has_four_waves` `anglo_egyptian_leaders_available_each_wave` `anglo_egyptian_turn_four_is_all_remaining` `campaign_reinforcement_cap_and_double_entry` `campaign_gunboats_quota_three_per_turn`

§9.211 – Anglo-Egyptian set up first, moves second (Historical)

`implemented`
manual page unknown

The Anglo-Egyptian player sets up first, and moves second:

- Not in play: GORDON leader unit, “Friendlies” brigade.
- Gunboats start in any Nile River hexes adjacent to the Zariba.
- Camel Corps, Egyptian Cavalry, and Horse Artillery start in the village of Kerreri hut hexes.
- All remaining Anglo-Egyptian units set up in the 13 hexes of the Zariba.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:2461 GH:omdurman-rules/src/effects.rs:2461	first_player	<pre> 2459 }); 2460 } 2461 2462 /// The player who moves first in a scenario (§4, §9.113, §9.212, §9.322). 2463 pub fn first_player(scenario: Scenario) -> Player { </pre>

Covered by tests: `historical_setup_rejects_not_in_play_units`

§9.212 – Dervish set up (Historical) – deployment zones and leader-to-hex mapping

`implemented`
manual page unknown

The Dervish player sets up second, and moves first.

- Not in play: Isa Zachneih, gunboats, and forts.
- All Dervish units must be set up out of the line of sight of all Anglo-Egyptian units.
- Dervish leaders start on the lettered hexes:
 - A: Ali Wad Helu
 - D: Sheik El Din
 - Y: Yakub
 - K: Khalifa Abdullah
 - S: Sherif
 - O: Osman Digna

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1007	in_deployment_zone	<pre> 1005 /// - **Historical / Campaign** (§9.211-9.212, §9.11): permissive. The 1006 /// manual's constraints there are the 13 Zariba </pre>

GH:omdurman-rules/src/effects.rs:1007		hexes, the Kerreri huts, 1007 /// and per-leader "within three hexes" color groups -- data the engine's 1008 /// `BoardInfo` does not carry (no Zariba-hex set, no Kerreri landmark, no 1009 /// per-unit leader color), so those are enforced by the scenario set-up
omdurman-rules/src/lib.rs:426 GH:omdurman-rules/src/lib.rs:426	Dervish- Leader::setup_letter	424 /// (§9.212): A→Ali Wad Helu, D→Sheik El Din, Y→Yakub, K→Khalifa Abdullah, 425 /// S→Sherif, O→Osman Digna. Inverse of [<code>dervish_leader_for_setup_letter</code>]. 426 pub fn setup_letter(self) -> SetupLetter { 427 match self { 428 DervishLeader::AliWadHelu => SetupLetter::A,
omdurman-rules/src/lib.rs:442 GH:omdurman-rules/src/lib.rs:442	dervish_leader_for_setup_l	440 /// inherent impl here, so the mapping is a free function -- the bijective 441 /// inverse of [<code>DervishLeader::setup_letter</code>]. 442 pub fn dervish_leader_for_setup_letter(letter: SetupLetter) -> DervishLeader { 443 match letter { 444 SetupLetter::A => DervishLeader::AliWadHelu,

Covered by
tests: `deploy_rejected_outside_zone` `embedded_leaders_resolve_from_their_host_section` `historical_places_all_six_leaders_when_anchors_present`
`missing_anchor_is_reported_not_dropped_silently` `setup_letter_dervish_leader_roundtrip` `setup_letter_to_dervish_leader_known_values`
`historical_setup_rejects_not_in_play_units`

§9.231 – Thorn hedge hexsides

`implemented`
manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:932 GH:omdurman-rules/src/lib.rs:932	ZaribaThornHedge	930 Terrain(i16), 931 /// -2 thorn-hedge defensive modifier (§9.231). 932 ZaribaThornHedge, 933 /// -4 trench defensive modifier (§9.232). Only applies vs. "entrenched" 934 /// units (those Nile-side of the trench hexside).
omdurman-types/src/lib.rs:195 GH:omdurman-types/src/lib.rs:195	ZaribaThornHedge	193 Crest, 194 /// Historical-scenario thorn-hedge segment of the Zariba (§9.231). 195 ZaribaThornHedge, 196 /// Historical-scenario trench segment of the Zariba (§9.232). 197 ZaribaTrench,

§9.232 – Trench hexsides

`implemented`
manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:935 GH:omdurman-rules/src/lib.rs:935	ZaribaTrenchEntrenched	933 /// -4 trench defensive modifier (§9.232). Only applies vs. "entrenched" 934 /// units (those Nile-side of the trench hexside). 935 ZaribaTrenchEntrenched, 936 } 937
omdurman-types/src/	ZaribaTrench	

lib.rs:197 GH:omdurman-types/src/lib.rs:197	<pre> 195 ZaribaThornHedge, 196 /// Historical-scenario trench segment of the Zariba (§9.232). 197 ZaribaTrench, 198 /// One of the two end hexsides of a Zariba trench segment that connect to 199 /// the Nile River (§9.233). Units may only enter/ leave the Zariba via </pre>
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§9.233 – Zariba entry/exit costs

implemented
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Units may only enter and/or leave the Zariba via the two end hexsides that connect to the Nile River, paying +2 movement points to cross (Exception: advance after combat across an entrenched hexside).

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:247 GH:omdurman-types/src/lib.rs:247	blocks_movement	<pre> 245 /// `omdurman-rules`). The trench *end* variants are therefore intentionally 246 /// not blocking. 247 pub fn blocks_movement(self) -> bool { 248 matches!(249 self, </pre>
omdurman-rules/src/board.rs:214 GH:omdurman-rules/src/board.rs:214	zariba_entry_surcharge	<pre> 212 /// movement points to cross"). Returns 2 when the edge between `from` and 213 /// `to` is one of the two trench ends, else 0. 214 pub fn zariba_entry_surcharge(&self, from: HexCoord, to: HexCoord) -> i16 { 215 match self.hexside_between(from, to) { 216 Some(k) if k.is_zariba_trench_end() => 2, </pre>
omdurman-rules/src/effects.rs:1385 GH:omdurman-rules/src/effects.rs:1385	movement_cost_for	<pre> 1383 /// falls back to its supplied cost). Land units pay each hex's terrain cost; 1384 /// gunboats pay one MP per Nile hex entered (§5.24 counts hexes, not 1385 /// terrain). The per-hex passability is enforced separately in the 1386 /// land/gunboat validators, so an off-map hex here contributes the clear- 1387 /// terrain base of 1. </pre>
omdurman-rules/src/effects.rs:2662 GH:omdurman-rules/src/effects.rs:2662	apply_move_unit	<pre> 2660 /// Validate and apply a unit movement (rulebook §5). When `path` is supplied 2661 /// (the entered hexes, excluding the start, ending at `to`) the engine computes 2662 /// the true terrain cost (§5.11) and enforces gunboat upstream/downstream 2663 /// allowances (§5.24); otherwise it falls back to the caller-supplied `cost`. 2664 pub fn apply_move_unit(</pre>

Covered by tests: [zariba_end_hexside_costs_extra_mp](#) [zariba_thorn_hedge_blocks_movement](#)

§9.321 – British set up (Bonus)

out-of-scope
manual page unknown

The British player sets up first, moves second:

- General GORDON leader unit in the palace.
- Two old style (unnamed) gunboats in any Nile River hexes.

- Set up in any building or hut hexes of Khartoum, Forts Makran and/or Buri, and/or adjacent to any wall hex:
 - one Egyptian Battalion artillery unit
 - two British infantry units (represents Caucasian troops)
 - three Egyptian infantry units (represents Cairo “Bazouks”)
 - four Sudan infantry units (represents Sudanese blacks)
 - four “Friendlies” units (represents the Shaggyeh)

(see manual for full text)

Covered by tests: `confirm_ready_rejected_below_scenario_target` `fall_of_khartoum_places_gordon_in_the_palace` `fall_of_khartoum_reports_missing_palace` `fall_of_khartoum_fort_landmarks_sit_at_the_correct_hexes` `fok_ae_gunboat_deploys_only_on_nile` `fok_ae_land_unit_rejected_on_nile` `fok_setup_complete_requires_full_oob` `deploy_via_real_sprite_resolution_matches_engine` `british_boats_named_vs_old_gunboat_detection`

§9.322 – Dervish enters turn one (Bonus)

implemented

manual page unknown

Dervish player moves first: enters turn one through any hexes on the south or east edge of the map.

- 32 Mulazmin units (represents combined forces of Wad El Nejumi, Abu Girgeh, and Sheik El Obeid)
- 2 Hadendowa; 6 Kehena; 5 Degheim (represents Mahdi’s combined west bank forces)
- 3 Dervish artillery units.

File	Symbol	Code Snippet
omdurman-rules/src/unit_profiles.rs:309 GH:omdurman-rules/src/unit_profiles.rs:309	<code>ali_wad_helu</code>	<pre> 307 /// (3-6-9) -- the Degheim force of §9.322, printed on Baggara-backed 308 /// sprites. 309 fn ali_wad_helu(col: u32, row: u32) -> Option<Classification> { 310 match (col, row) { 311 (0, 0) => dervish_leader(DervishLeader::AliWadHelu), </pre>
omdurman-types/src/lib.rs:777 GH:omdurman-types/src/lib.rs:777	<code>sections_for_picker</code>	<pre> 775 /// provides the 3 artillery, and HadendowaForts supplies the 776 /// Dervish-controlled North Fort sprite (§9.344). 777 pub fn sections_for_picker(self) -> Option<&'static [SectionName]> { 778 match self { 779 Scenario::Campaign Scenario::Historical => None, </pre>

Covered by tests: `ali_wad_helu_block_resolves_leader_and_degelim_tribes` `fok_setup_complete_requires_full_oob` `fok_dervish_land_unit_rejected_on_nile` `fok_picker_allowlist_has_dervish_entry_force_blocks`

§9.341 – Turn 1 is always a night turn (Bonus)

implemented

manual page unknown

Turn 1 is always a night turn (see 8.1).

See also: §8.1

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:172 GH:omdurman-	<code>FALL_OF_KHARTOUM</code>	<pre> 170 /// (the rulebook fixes none); only `day_night` is rule-bearing (night halves 171 /// Anglo-Egyptian movement and ranges and bars </pre>

rules/src/ turn_track.rs:172		howitzer fire, §8.1). 172 pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] = [173 TurnEntry { 174 turn: 1,
---------------------------------	--	--

Covered by tests: [fall_of_khartoum_turn_one_is_night](#)

§9.342 – All hexes are playable, including half hexes (Bonus)

[implemented](#)

manual page unknown

All hexes are playable, including hexes showing up half or less.

File	Symbol	Code Snippet
omdurman-rules/src/board_data.rs:2006 GH:omdurman-rules/src/ board_data.rs:2006	fall_of_khartoum_map_data	2004 /// §9.31). Every hex on the board is playable, including the half-hexes along 2005 /// the map edges (rulebook §9.342) -- there is no `excluded` set. 2006 pub fn fall_of_khartoum_map_data() -> MapData { 2007 let tiles = { 2008 let mut _m = BTreeMap::new();

§9.343 – Both players use the Dervish Range Effects Table (Bonus)

[implemented](#)

manual page unknown

Both players must use the Dervish Range Effects Table.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:2782 GH:omdurman-rules/src/ effects.rs:2782	range_band_for	2780 /// Look up the range-effects band for a firing unit. Normally Anglo-Egyptian 2781 /// units use their own table and Dervish units the Dervish table (§6.22), but 2782 /// in FALL OF KHARTOUM *both* players use the Dervish Range Effects Table 2783 /// (§9.343). 2784 pub fn range_band_for(

§9.344 – Dervish controls the North Fort (Bonus)

[implemented](#)

manual page unknown

The Dervish player controls the “North Fort” and may fire its guns.

File	Symbol	Code Snippet
omdurman-app/src/scenario_setup.rs:115 GH:omdurman-app/src/scenario_setup.rs:115	FALL_OF_KHARTOUM	113 /// GORDON is the "GEN. GORDON" counter at British_Boats (3,1); the North Fort 114 /// uses a campaign HadendowaForts counter (one of the spare fort sprites). 115 const FALL_OF_KHARTOUM_SETUP: &[FixedPlacement] = &[116 FixedPlacement { 117 section: SectionName::BritishBoats,
omdurman-rules/src/	hex_has_enemy_fort	

effects.rs:1928 GH:omdurman-rules/src/effects.rs:1928	<pre> 1926 1927 /// Whether `hex` holds a fort owned by `mover`'s 1928 /// enemy. Per §6.54 a player 1929 /// may neither occupy an enemy fort nor advance 1930 /// after combat into one 1931 /// (forts are never captured -- only destroyed, 1932 /// §6.62/§6.53/§7.6). 1933 pub fn hex_has_enemy_fort(&self, hex: HexCoord, 1934 mover: Player) -> bool { </pre>
--	--

Covered by tests: [fall_of_khartoum_places_gordon_in_the_palace](#) [fall_of_khartoum_fort_landmarks_sit_at_the_correct_hexes](#) [placement_done_gate_matches_by_identity_not_allocated_id](#)

§9.345 – Gunboat White Nile <-> Blue Nile crossing (Bonus)

[implemented](#)

manual page unknown

The British gunboats may move from the White Nile to the Blue Nile and vice-versa at an off-board movement cost of six “upstream” movement points.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1912 GH:omdurman-rules/src/effects.rs:1912	is_nile_mouth_crossing	<pre> 1910 /// Whether moving from `from` to `to` is the 1911 /// §9.345 off-board crossing 1912 /// between the two Nile-branch mouths (in either 1913 /// direction). Both mouths 1914 /// must be named on the board, else this is 1915 /// `false` and the move falls 1916 /// through to the ordinary contiguous-Nile rules. 1917 pub fn is_nile_mouth_crossing(&self, from: 1918 HexCoord, to: HexCoord) -> bool { </pre>
omdurman-types/src/lib.rs:525 GH:omdurman-types/src/lib.rs:525	Location::WhiteNileMouth	<pre> 523 /// The off-board mouth of the White Nile branch 524 /// (FALL OF KHARTOUM §9.345) -- 525 /// a British gunboat may cross to the Blue Nile 526 /// mouth for 6 upstream MP. 527 WhiteNileMouth, 528 /// The off-board mouth of the Blue Nile branch 529 /// (FALL OF KHARTOUM §9.345). 530 BlueNileMouth, </pre>

§9.346 – GORDON immobile, eliminated only at the Palace (Bonus)

[implemented](#)

manual page unknown

The GORDON leader unit starts in the palace and may not move during the scenario. He may only be eliminated by a Dervish unit passing through or occupying the palace hex (as normal movement or as advance after combat).

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:2590 GH:omdurman-rules/src/effects.rs:2590	check_gordon_palace	<pre> 2588 /// §9.346: in FALL OF KHARTOUM, GORDON is eliminated 2589 /// the instant a Dervish unit 2590 /// passes through or occupies the Palace hex (by 2591 /// normal movement or advance 2592 /// after combat). Records the turn (which fixes the 2593 /// §9.35 victory level) and 2594 /// ends the game. A no-op outside FoK, or once GORDON 2595 /// is already gone. 2596 pub fn check_gordon_palace(state: &mut GameState) { </pre>
omdurman-rules/src/lib.rs:662	UnitIdentity::is_gordon	<pre> 660 /// Whether this is the GORDON leader unit (§9.32, 661 /// §9.346) -- the immobile 662 /// palace defender whose elimination ends FALL OF </pre>

GH:omdurman-rules/src/lib.rs:662	<pre>KHARTOUM (\$9.35). 662 pub fn is_gordon(&self) -> bool { 663 matches!(664 self,</pre>
----------------------------------	--

Covered by tests:

fall_of_khartoum_places_gordon_in_the_palace

gordon_is_an_immobile_british_leader

gordon_survives_means_no_elimination

 7/10 implemented

§10 – Optional Rules

§10 – Optional Rules

implemented

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Optional Rules (Campaign game only)

It is suggested that the most intriguing employment of the following two options is to permit the Dervish player to have either one or the other, but the Anglo-Egyptian player doesn't know which one until he stumbles onto it. Players are advised that the employment of both optionals in the same game is not recommended.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:355 GH:omdurman-rules/src/lib.rs:355	OptionalRule	<pre>353 /// two should be in play (rulebook §10). 354 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 355 pub enum OptionalRule { 356 RiverMines, 357 RiverChain,</pre>

§10.1 – River Mines

implemented

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River Mines

The Khalifa twice tried (unsuccessfully) to submerge a powerful mine in the Nile to sink or damage British gunboats. This option assumes that both attempts were successful.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:145 GH:omdurman-rules/src/effects.rs:145	RiverMine	<pre>143 // -- Optional rules ----- 144 /// River mine resolution (rulebook §10.12). 145 RiverMine { 146 gunboat_id: UnitId, 147 hex: HexCoord,</pre>
omdurman-rules/src/effects.rs:4294 GH:omdurman-rules/src/effects.rs:4294	apply_river_mine	<pre>4292 } 4293 } 4294 Ok(()) 4295 } 4296 </pre>
omdurman-rules/src/lib.rs:1116 GH:omdurman-rules/src/lib.rs:1116	MineResult	<pre>1114 /// British gunboat enters a mined hex. 1115 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1116 pub enum MineResult { 1117 /// Roll 1-4: no effect. 1118 NoEffect,</pre>

§10.2 – River Chain

out-of-scope

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River Chain

The Khalifa also tried (also unsuccessfully) to string a heavy chain across the Nile to stop or slow down the British gunboats. This option assumes the chain was emplaced.

§10.11 – Secretly record mine hexes

out-of-scope
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Prior to the commencement of play the Dervish player secretly records two Nile River hexes to be mined (the mines may not both be placed in the same hex). These hexes must be south of the E–W hexrow in which the Khor Shambat empties into the Nile.

Covered by tests: mine_and_chain_limits_enforced_in_setup mines_and_chain_require_their_optional_rule

§10.12 – Mine resolution

implemented
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When a British gunboat enters a mined hex, the Dervish player must order it to stop as it has struck a mine. The Dervish player then resolves the effect of the mine’s blast by rolling the ten-sided die:

- 1–4: No effect
- 5–7: Gunboat damaged, lost use of its engines and must drift two hexes per turn (with the current) for the rest of the game. No effect on guns or Maxims unless they drift out of range.
- 8–10: Gunboat sunk!

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:145 GH:omdurman-rules/src/effects.rs:145	RiverMine	<pre>143 // -- Optional rules ----- 144 /// River mine resolution (rulebook §10.12). 145 RiverMine { 146 gunboat_id: UnitId, 147 hex: HexCoord,</pre>
omdurman-rules/src/effects.rs:4294 GH:omdurman-rules/src/effects.rs:4294	apply_river_mine	<pre>4292 } 4293 } 4294 Ok(()) 4295 } 4296 </pre>

§10.13 – Mines consumed after both rolled for

implemented
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After both mines have been rolled for, no more are available.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:4294 GH:omdurman-rules/src/effects.rs:4294	apply_river_mine	<pre>4292 } 4293 } 4294 Ok(()) 4295 } 4296 </pre>

§10.14 – Dervish gunboats pass safely

implemented

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The Dervish player's gunboats may pass through the mined hexes with no ill effect (he knows where the mines are).

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:4294 GH:omdurman-rules/src/effects.rs:4294	<code>apply_river_mine</code>	<pre>4292 } 4293 } 4294 Ok(()) 4295 } 4296 </pre>

§10.21 – Secretly record chain hexes

out-of-scope

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Prior to the commencement of play the Dervish player secretly records a line of river hexes (not exceeding four hexes long) across which the chain is strung. The hexes must be south of the E–W hexrow in which the Khor Shambat empties into the Nile.

Covered by tests: [mine_and_chain_limits_enforced_in_setup](#) [mines_and_chain_require_their_optional_rule](#)

§10.22 – Gunboat stops on chained hex

implemented

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When a British gunboat enters a “chained” river hex it must stop and may move no further that turn.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1418 GH:omdurman-rules/src/effects.rs:1418	<code>can_move_gunboat</code>	<pre>1416 } 1417 1418 /// Validate a gunboat move along `path` (§5.22, 1419 /// move only along Nile hexes; their two 1420 /// and downstream (larger); and "if they move even one hex upstream, their</pre>

§10.23 – Sinking the chain

implemented

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No gunboats (British or Dervish) may cross the chain until it has been sunk by the British player. He may sink the chain by a) having an infantry or cavalry unit spend one complete turn on either riverbank adjacent to a “chained” river hex, or b) firing at the chain with artillery and achieving a 3 or more on the Combat Results Table.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:4338 GH:omdurman-	<code>apply_sink_chain</code>	<pre>4336 // Move the gunboat downstream. Gunboats ignore 4337 stacking (§5.51). 4338 if let Some(u) = state.find_unit_mut(unit_id) { 4339 u.position = downstream; }</pre>

rules/src/ effects.rs:4338		4340 // If the gunboat drifts into a mine, resolve it immediately.
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0/1 implemented

§11 – Historical Notes

§11 – Historical Notes

descriptive

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Historical Notes

In 1881 Mohammed Ahmed Ibn Al-Sayid Abdullah, the son of an obscure carpenter in the hinterlands of the Sudan, proclaimed himself the “Mahdi” — the Messiah of the Islamic faith. His timing was propitious indeed. Since the early 1820’s a corrupt Egypt, with the Sultan of Turkey’s blessing, had incessantly raped the Sudan, taking ivory and flooding the slave markets with some half million captured Sudanese blacks. By 1880, nearly 40,000 Egyptian troops occupied outposts scattered throughout the Sudan, enforcing Egypt’s hold on this lucrative ivory and slave trade and squeezing the native population dry through vicious and corrupt tax officials. All was controlled from Khartoum via the office of Governor General of the Sudan. The title had been held by a succession of individuals, including General Charles Gordon, whose appointment was an attempt to reinstate some rudimentary justice in the Sudan after France and Britain assumed joint political control of a bankrupt Egypt.

By 1881, however, Gordon’s term had expired and a new Governor General, again corrupt and incompetent,
(see manual for full text)

See also: §9.1, §9.2, §9.3

0/1 implemented

Credits

§Credits – Credits

[descriptive](#)

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 1/1 implemented

Combat Results Table (shared reference)

§CRT – Combat Results Table (shared by §6.22 fire and §7.7 melee)

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/combat_results_table.rs:8 GH:omdurman-rules/src/combat_results_table.rs:8	FireFactorRow	<pre>6 /// to index into the result matrix. 7 #[derive(serde::Serialize, serde::Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 8 pub enum FireFactorRow { 9 /// 1-5 factors 10 Row01to05,</pre>
omdurman-rules/src/combat_results_table.rs:31 GH:omdurman-rules/src/combat_results_table.rs:31	from_total	<pre>29 impl FireFactorRow { 30 /// Determine which row a given total fire factor falls into (rulebook §6.22). 31 pub fn from_total(total: u16) -> Self { 32 match total { 33 0..=5 => FireFactorRow::Row01to05,</pre>
omdurman-rules/src/combat_results_table.rs:69 GH:omdurman-rules/src/combat_results_table.rs:69	combat_results_table	<pre>67 /// D = `Disrupt` (1/2 of target units, round up) 68 /// 1...5 = `Eliminate(n)` (that many units removed) 69 pub fn combat_results_table(row: FireFactorRow, roll: DieRoll) -> CombatResult { 70 use CombatResult::*; 71 use DieRoll::*;</pre>
omdurman-rules/src/lib.rs:996 GH:omdurman-rules/src/lib.rs:996	CombatResult	<pre>994 /// * `--` -- no effect 995 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 996 pub enum CombatResult { 997 NoEffect, 998 Disrupt,</pre>

Covered by tests: [ae_combat_results_table_lowest_is_no_effect](#)
[ae_combat_results_table_highest_is_eliminate_5](#) [ae_combat_results_table_progresses_with_roll](#) [ae_combat_results_table_progresses_with_factor](#)
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[crt_eliminate_never_exceeds_5](#) [crt_every_cell_matches_the_table](#)

0/1 implemented

Reference – Charts and Tables

§Reference – Charts and Tables

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